Solving Nonlinear Schrödinger Equations using Darboux Transformation Method

by

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Abstract

The study of colour superconducting matter within the core of compact stars is an active area of research. This thesis studies the thermodynamics of two-flavour colour superconductivity and its effects on the structure and cooling of compact stars containing matter in that phase. The colour-superconducting phases studied include the pure two-flavour colour superconductors (2SC), gapless 2SC (g2SC), mixed phase of non-superconducting (normal) quark matter and 2SC or g2SC phase, with possibly free strange quarks present.

We employ the two-flavour SU(2) Nambu-Jona-Lasinio model for the colour-superconducting phase. For modelling the low density matter on the outer region of compact stars, we solve the neutral hadronic phase using the relativistic mean field method. The equation of state of the colour superconducting phases and the hadronic phase are found, as well as the phase diagram of the quark matter. The quark matter is seen to undergo second order phase transitions from the mixed phase to pure 2SC or g2SC phase and in between the 2SC and g2SC phases. Without considering the mixed phase, the quark matter also undergoes a second order phase transition from normal to superconducting phase with increase of the baryon chemical potential.

To find the mass and structure of the compact stars, we solve the general relativistic equations of hydrostatic equilibrium together with the equation of state of matter in the particular phase of interest. With the change of the bag constant used, we find a transition from pure quark to hybrid and ultimately pure hadronic stars. For certain values of the bag constant, with the increase of central pressure, we see an unstable branch in the mass versus radius plot of the hybrid stars which later changes into a stable branch. We find the difference of mass and total baryon number between configurations of the compact stars at zero and finite central temperatures as functions of the central pressure. It is seen that the mass and total baryon number differences have maxima at certain initial central pressures for each final finite temperature.

The cooling stars are considered at the fixed total baryon number as

that of the initial finite-temperature star. Depending on the initial central pressure, the central pressure of the star may either increase or decrease during cooling. We find the thermal capacity or the ratio of the energy lost to the change of central temperature for quark stars in the 2SC/g2SC or mixed phase with and without free strange quarks. The quark stars without free strange quarks in 2SC phase are seen to lose more energy as they cool as compared to stars with free strange quarks. The cooling quark stars are however seen to shrink more if they contain free strange quarks.

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Chapter 1

Introduction

The nature is very beautiful, strange and sometimes so much astonishing that it is always attracting us and drawing our attention to discover its secrets, to decipher what is encoded into it.And we human are always curious about decoding this nature.We start this decoding process by questioning "Why is it happening? Why is it not happening?".Then, what we do is that we try to find a easy way to explain our nature and solve the problems of nature that is harmful for the living being.The easiest way to explain any natural phenomena is finding an equation that can explain it.
Many scientists with their extraneous labour have been trying to do this.The great scientist Albert Einstein once said about equation that "One has to divide one's time between politics and our equations.Equations are much more important to me, because politics is for the present, while an equation is for eternity."In this thesis we will discuss about a kind of equation, which is not simple like linear equation.So, we are going to step up to the world of non-linear equations.We will try to solve a nonlinear equation called nonlinear Schödinger equation by using the simplest Darboux Transformation(DT) Method .Our title of this thesis is " Solving Nonlinear Schrödinger Equations using Darboux Transformation Method"

1.1 Objective

The nonlinear Schrödinger equation (NLSE) appears in different branches of physics. In optical fiber communication systems, this type of equation describes the dynamics of an optical pulse through the fiber. Due to the advent of high power optical soliton, higher-order terms become very important in describing the soliton propagation accurately through the fiber. The role of the higher-order terms in the stability and dynamics of the solitons thus needs to be understood. To that end, we consider here the most general

form of the NLSE with cubic non-linearity, complex potential, and derivative terms. There are many different approaches to solve the nonlinear partial differential equations, some of which are developed very recently.

There are many methods for solving nonlinear Schrödinger equations like Hirota bilinear method, the generalized and improved expansion method, the mapping method, and Darboux Transformation (DT) method.We solve this general equation with the DT method. It has the beauty of generating non-trivial solutions from a trivial seed solution of a given integrable NLSE.We discuss the integrability conditions and solve some of the well-known NLSEs with cubic non-linearity and time dependent coefficients. The effect of the higher-order terms will then be evident. In some cases, we include a linear potential and a term describing the damping or gain in the system.

The Darboux transformation is one of the most important tools for generating the solution of an integrable cases of quantum mechanical equations. Originally developed as a mathematical transformation that relates solutions of ordinary diferential equations to each other. in the context of quantum mechanics the Darboux transformation was first found to be applicable to the linear stationary Schrödinger(NLSE) equation Ever since then, Darboux transformations have been constructed for a variety of linear and nonlinear equations.

1.2 The Aim

The aim of the paper is to use the Darboux transformation method to get exact solitonic solutions for higher order nonlinear Schrödinger equation HNLSE with complex potentials, and time-dependent coefficients cubic and quintic non-linearities together and see also the effect of the higher order terms.

If we can understand the behaviour of the solitons(e.g. water waves and optical waves) we will be able to describe them. We will be able to know how to control them.Then one day, the natural disasters like tsunami, dangerous tidal waves all can be predicted and we can save the living beings

from the curse of them.
Now a days, by applying the nonlinear effect in optical fiber we can develop the laser technology, The need of communication is an all time need of human beings. For communication some channel is needed. Fiber is one channel among many other channels for communication. The dispersion phenomenon is a problem for high bit rate and long haul optical communication systems. An easy solution of this problem is optical soliton pulses that preserve their shape over long distances. Soliton based optical communication systems can be used over distances of several thousands of kilometers with huge information carrying capacity by using optical amplifiers[23].

1.3 Methods of study

Our approach of finding the solution was based on analytical approach.We used the programing language Mathematica.Here, first we showed the matrices and simplified the NLSE for those matrices.We found the Lax pairs and seed solutions.Then we used Darboux Transformation method on the seed solution.At the end of the code we have got our expected nontrivial solution.After changing parameters we have got different solutions.The solution changed because of changing the seed solution.For finding the Lax pair we had to use higly efficient sofwares.

1.4 Organization of Chapters

Apart from the introductory chapter,this thesis contains four other chapters.In second chapter we described different types of Schrödinger equations(linear and non-linear) and different types of solutions to the Nonlinear Schrödinger equations(NLSE), like numerical and analytical solutions.In third chapter we defined Darboux DT, DT theorem, how to use DT,its applications, DT for nonstationary SE, Lax pair,Lax equation, Its isospectral properties etc.In the forth chapter we did result discussions.In the fifth chapter, finally we summarized our findings and further investigation.

Chapter 2

Solution techniques for Schrödinger equations

The Schrödinger equation is the fundamental equation of physics for describing quantum mechanical behavior. It is also often called the Schrödinger wave equation, and is a partial differential equation that describes how the wave function of a physical system evolves over time. Viewing quantum mechanical systems as solutions to the Schrödinger equation is sometimes known as the Schrödinger picture, as distinguished from the matrix mechanical viewpoint, sometimes known as the Heisenberg picture. [5]

The Schrodinger equation plays the role of Newton's laws and conservation of energy in classical mechanics - i.e., it predicts the future behavior of a dynamic system. It is a wave equation in terms of the wave function which predicts analytically and precisely the probability of events or outcome. The detailed outcome is not strictly determined, but given a large number of events, the Schrödinger equation will predict the distribution of results.[27]

2.1 Types of Schrödinger equation:

On the basis of time dependence Schrödinger equation can both be time independent and time dependent.

1. Time independent:The time-independent Schrödinger equation predicts that wave functions can form standing waves, called stationary states.The most general form is the time-dependent Schrödinger equation can be written as:

$$E\psi(x) = [-\frac{\hbar^2}{2m}\nabla^2 + V(x)]\psi(x)$$

2. Time dependent:The form of the Schrödinger equation depends on the physical situations. The most general form is the time-dependent

Schrödinger equation, which gives a description of a system evolving with time:

$$i\hbar\frac{\partial}{\partial t}\psi(x,t) = H\psi(x,t)$$

The most famous example is the non-relativistic time dependent Schrödinger equation for a single particle moving in an electric field can be written as:

$$i\hbar\frac{\partial}{\partial t}\psi(x,t) = [-\frac{\hbar^2}{2m}\nabla^2 + V(x,t)]\psi(x,t)$$

On the basis of linearity and non-linearity Schrödinger equation is of two types.[25]

- Linear Schrödinger equation
- Nonlinear Schrödinger equation

2.2 Linear Schrödinger equation:

In theoretical physics, the (one-dimensional) linear Schrödinger equation (NLSE) is a linear variation of the Schrödinger equation.

2.2.1 Example of Linear Schrödinger equation

Schrödinger Equation for Quantum Harmonic Oscillator is an important example of linear Schrödinger equation. The Schrödinger equation for a harmonic oscillator may be obtained by using the classical spring potential,
V(x)=$\frac{1}{2}kx^2$=$\frac{1}{2}m\omega^2x^2$
ω = angular frequency
ω=2π.frequency
The time independent Schrödinger equation with this form of potential is,

$$-\frac{\hbar^2}{2m}\frac{\partial^2\psi}{\partial x^2} + \frac{1}{2}m\omega^2x^2\psi = E\psi \tag{2.1}$$

Here,

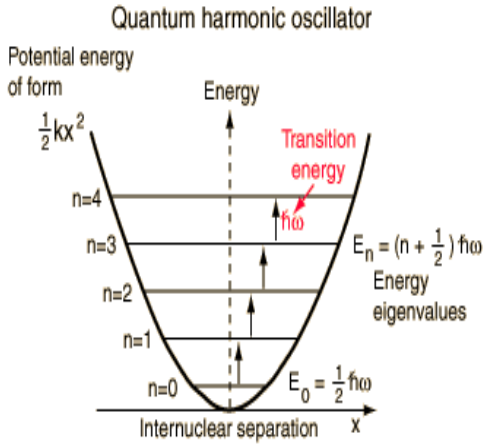

Figure 2.1: Energy eigenvalues for linear quantum harmonic oscillator.

The form of Hamiltonian operator for a quantum harmonic oscillator, H=$-\frac{\hbar^2}{2m}\frac{\partial^2\psi}{\partial x^2}+\frac{1}{2}m\omega^2x^2\psi$
Kinetic Energy=$-\frac{\hbar^2}{2m}\frac{\partial^2}{\partial x^2}$
Potential Energy=$\frac{1}{2}m\omega^2x^2$
Energy eigenvalue for the the quantum harmonic oscillator are given by,$E_n = (n + \frac{1}{2})\hbar\omega$ n=1,2,3,...
To obtain specific values for energy, you operate on the wave function with the quantum mechanical operator associated with energy, which is called the Hamiltonian.The operation of the Hamiltonian on the wave function is the Schrödinger equation. Solutions exist for the time-independent Schrödinger equation only for certain values of energy, and these values are called "eigenvalues" of energy[Figure-2.1].

The time dependent Schrödinger equation is,

$$-i\frac{\partial}{\partial t}\psi = -\frac{1}{2}\frac{\partial^2\psi}{\partial x^2} + \frac{1}{2}x^2\psi \tag{2.2}$$

We can derive equation (1.2) from equation (1.1) by changing the scaling.

In equation (1.1), $\psi \longrightarrow$wave function
Total energy,E=i$\hbar\frac{\partial}{\partial t}$

From equation (1.1) we get.

$$-\frac{\hbar^2}{2m}\frac{\partial^2\psi}{\partial x^2}+\frac{1}{2}m\omega^2x^2\psi=i\hbar\frac{\partial}{\partial t}\psi$$

Dividing both side by $\hbar\omega$ we get,

$$\begin{aligned}
\Rightarrow \frac{i\hbar}{\hbar\omega}\frac{\partial\psi}{\partial t} &= \frac{\hbar^2}{2\hbar m\omega}\frac{\partial^2\psi}{\partial x^2}+\frac{m\omega^2x^2}{2\hbar\omega}\psi \\
\Rightarrow \frac{i}{\omega}\frac{\partial\psi}{\partial t} &= \frac{\hbar}{2\hbar m\omega}\frac{\partial^2\psi}{\partial x^2}+\frac{m\omega x^2}{2\hbar}\psi \\
\Rightarrow -i\frac{\partial}{\partial(t\omega)}\psi &= -\frac{1}{2}\frac{\hbar}{m\omega}\frac{\partial^2\psi}{\partial x^2}+\frac{m\omega}{\hbar}\frac{x^2}{2}\psi \\
\Rightarrow -i\frac{\partial\psi}{\partial t'} &= -\frac{1}{2}{a_x}^2\frac{\partial^2\psi}{\partial x^2}+\frac{1}{2}\frac{1}{{a_x}^2}x^2\psi \\
&taking \\
&t'=t\omega \\
&\frac{\hbar}{m\omega}={a_x}^2 \\
\Rightarrow -i\frac{\partial\psi}{\partial t'} &= -\frac{1}{2}{a_x}^2\frac{\partial^2\psi}{{a_x}^2\partial x'^2}+\frac{1}{2}\frac{{a_x}^2x'^2}{{a_x}^2}\psi \quad [x\longrightarrow a_{xx'}] \\
\Rightarrow -i\frac{\partial\psi}{\partial t'} &= -\frac{1}{2}\frac{\partial^2\psi}{\partial x'^2}+\frac{x'^2}{2}\psi \\
\Rightarrow -i\partial_{t'}\psi &= -\frac{1}{2}\partial x x'\psi+-\frac{1}{2}x'^2\psi
\end{aligned}$$

As t' and x' are dummy indexes , this equation is exactly similar to equation (1.2)[The time dependent linear Schrödinger equation for harmonic oscillator]. In this way any time independent Schrödinger equation can be converted to time dependent equation.

2.3 Nonlinear Schrödinger equation

In theoretical physics, the (one-dimensional) nonlinear Schrödinger equation (NLSE) is a nonlinear variation of the Schrödinger equation.. It is a classical

field equation whose principal applications are to the propagation of light in nonlinear optical fibers and planar waveguides and to Bose-Einstein condensates confined to highly anisotropic cigar-shaped traps, in the mean-field regime.
In quantum mechanics, the 1D NLSE is a special case of the classical nonlinear Schrödinger field, which in turn is a classical limit of a quantum Schrödinger field.The nonlinear Schrödinger equation is an example of a universal nonlinear model that describes many physical nonlinear systems.

The nonlinear Schrödinger equation (NLSE) plays a vital role in different fields of science, e.g., physics, chemistry, biology, etc. In physics, NLSEs are very important class of equations and constitute an active field of research specially for the optical communication, Bose-Einstein condensation, and ocean waves communities. NLSEs describe the dynamics of optical pulses in optical fibers and photonic crystals, matter waves in Bose-Einstein condensates, and solitary and rogue waves in the ocean.The equation can also be applied to hydrodynamics,nonlinear optics,nonlinear acoustics, quantum condensates, heat pulses in solids and various other nonlinear instability phenomena.

2.3.1 Nonlinear Schröodinger equation with constant coefficients

The Nonlinear Schrödinger equation (NLSE) or the homogeneous Gross-Pitaevskiiequation (GPE) can be written as
$i\psi_t + a_1\psi_{xx} + a_2|\psi|^2\psi = 0$,.....(a)
where a_1 and a_2 represent the chromatic dispersion and non-p0linearity, respectively, and $\psi \equiv \psi(x,t)$. The subscript t and x denote the derivative with respect to time and position, respectively. In this section, we assume that both the co-efficients a_1 and a_2 are constants. This is one of the most well-known NLSEs in the optical [8], Bose-Einstein [24], and ocean wave communities [19]. Equation (a) is integrable, i.e., it has analytical solutions, for arbitrary constant values of of a_1 and a_2.

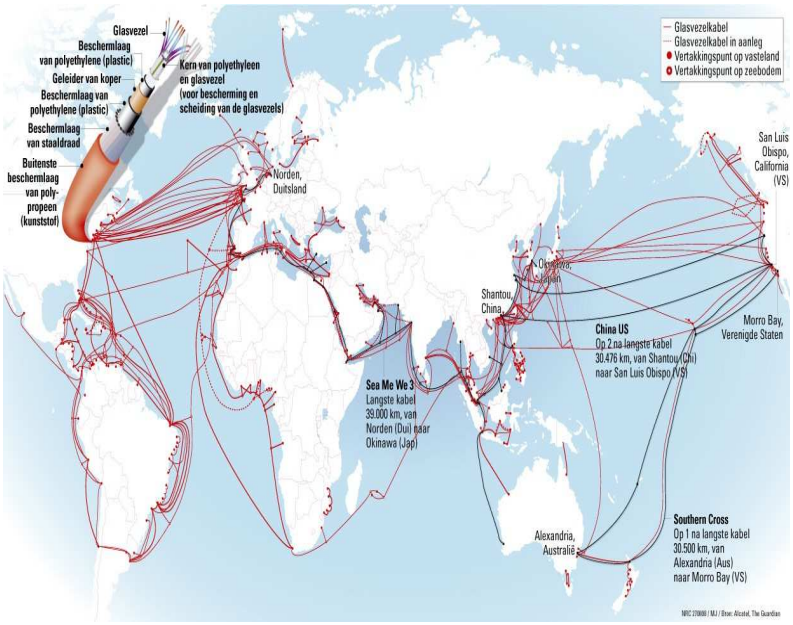

Figure 2.2: Optical fibers carry informations for hundreds, or even thousands, of kilometers.

2.3.2 Examples of Nonlinear Schrödinger equations(NLSE) in different fields

The nonlinear Schrödinger equation in fiber optics

[13] In optics, the nonlinear Schrödinger equation occurs in the Manakov system, a model of wave propagation in fiber optics. The function ψ represents a wave and the nonlinear Schrödinger equation describes the propagation of the wave through a nonlinear medium. The second-order derivative represents the dispersion, while the k term represents the non-linearity. The equation models many non-linearity effects in a fiber, including but not limited to self-phase modulation, four-wave mixing, second harmonic generation, stimulated Raman scattering, etc. [27]

Optical fibers made of pure glass (silica) are used as a medium for telecommunication and networking as they act as waveguides to transmit light between the two fiber ends. The information carrying signal travels for hundreds, or even thousands, of kilometers and undergoes several degeneration

Figure 2.3: Noise is mixed into the signal as the signal is reinforced at every amplifier along its path.

processes. Further undesired noise enters the system due to the fact that the signal needs to be periodically reinforced by optical amplifiers over its long journey.Modeling and simulation of the signal and the degeneration processes is fundamental for the correct interpretation of the output signal and is,therefore, of central interest for the telecommunications industries. Degeneration processes include chromatic dispersion (different wavelength travel with different speed), nonlinear self-phase modulation (Kerr effect) and dissipation. Furthermore the signal and noise is non-linearly mix up during the propagation, and this contributes to the degeneration of the output. All these phenomena can be described by a one dimensional, nonlinear Schrodinger equation, with stochastic input data.[5]

The nonlinear Schrödinger equation in water waves

For water waves, the nonlinear Schrödinger equation describes the evolution of the envelope of modulated wave groups. In a paper in 1968, Vladimir E. Zakharov describes the Hamiltonian structure of water waves. In the same paper Zakharov shows, that for slowly modulated wave groups, the wave amplitude satisfies the nonlinear Schrödinger equation, approximately.[2] The value of the non-linearity parameter k depends on the relative water depth. For deep water, with the water depth large compared to the wave length of the water waves, k is negative and envelope solitons may occur.

For shallow water, with wavelengths longer than 4.6 times the water depth, the non-linearity parameter k is positive and wave groups with envelope solitons do not exist. Note, that in shallow water surface-elevation solitons or waves of translation do exist, but they are not governed by the nonlinear Schrödinger equation.

The nonlinear Schrödinger equation is thought to be important for explaining the formation of rogue waves [13].

2.3.3 Special types of Nonlinear Schrödinger Equations

Hirota Equation (HE):

The general Hirota equation (HE) can be written as
$i\psi_t + a_1\psi_x x + a_2|\psi|^2\psi + ia_3\psi_x xx + ia4|\psi|^2\psi_x + i\gamma\psi + V_{ext}\psi = 0,$

which supports different type of solutions.

2.3.4 Solitonic solution:

This equation is integrable with $a_4 = 3a_2a_3/a_1$, $\gamma = (\dot{a}_4/a_4 - \dot{a}_3/a_3)/2$, and $V_ext = 0$, and the solution is given by
$\psi(x,t) = \frac{A(t)}{cosh[\sqrt{6}x - b(t)]} e^{6i\int a_1 dt}$

where A(t) $= -12\sqrt{a_3/a_4} and b(t) = 6\sqrt{6}\int a_3 dt$

2.3.5 Sasa-Satsuma Equation (SSE):

The general Sasa-Satsuma equation (SSE) can be written as

$i\psi_t + ia_3\psi_x xx + ia_4|\psi|^2\psi_x + ia_5\psi^2{\psi_x}^* + i\gamma\psi + V_e xt\psi = 0,$

which supports different type of solutions [8].

The solutions of NLSEs are thus significant in these different fields.

2.4 Solutions to the nonlinear Schrödinger equation

The examination of exact solutions of nonlinear evolution equations (NLEEs),like nonlinear Schrödinger's equations have a very important place in the inquiry

of some physical phenomena. The types of solutions of NLEEs, that are combined utilizing variety mathematical techniques, are very significant various sciences such as chemistry, technology of space, control engineering problems, physics, applied mathematics and computer engineering. In this paper, the nonlinear Schrödinger's equation (NLSE) that arises in the study of long-distance optical communications and all-optical ultra fast switching devices will be investigated. Some solutions for this equation can be found There are many different approaches to solve the nonlinear Schrödinger equation, some of which are developed very recently.
These solutions are divided into two parts [4].

1. Numerical approaches or Numerical solutions
2. Analytical approaches or Analytical solutions

Suppose you have a mathematical model and you want to understand its behavior. That is, you want to find a solution to the set of equations. The best is when you can use calculus, trigonometry, and other math techniques to write down the solution. Now you know absolutely how the model will behave under any circumstances. This is called the analytic solution, because you used analysis to figure it out. It is also referred to as a closed form solution.

But this tends to work only for simple models. For more complex models, the math becomes much too complicated. Then you turn to numerical methods of solving the equations, such as the Runge-Kutta method. For a differential equation that describes behavior over time, the numerical method starts with the initial values of the variables, and then uses the equations to figure out the changes in these variables over a very brief time period. Its only an approximation, but it can be a very good approximation under certain circumstances.

A computer must be used to perform the thousands of repetitive calculations involved. The result is a long list of numbers, not an equation. This long list of numbers can be used to drive an animated simulation, as we do with the models presented here.

There is also a middle ground between these two methods. There are many important non-linear equations for which it is not possible to find an analytic solution. However, there are techniques where you can find approximate analytic solutions that are close to the true solution, at least within a certain range. One such method is called the perturbation method. The advantage over a numerical solution is that you wind up with an equation (instead of just a long list of numbers) which you can gain some insight from.

Analytical is exact, numerical is approximate. In general, analytical solutions to the full Maxwell wave equation for a nonlinear optical system do not exist.Even numerical solutions to the wave equation are extremely difficult to implement due to the dimensionality of the problem.

2.4.1 Numerical Solutions

Various numerical methods are employed in order to approximate the nonlinear Schrödinger equation.The following numerical methods were applied to the NLS equation.

Finite difference methods:

- Explicit methods..
    1. The classical explicit method.
    2. The hopscotch method.
- Implicit methods.
    1. Implicit for the linear part and explicit for the nonlinear part (implicit-explicit).
    2. Crank-Nicolson implicit scheme.
    3. The Ablowitz and Ladik scheme.
- Finite Fourier transform or pseudo-spectral methods:
    1. Split step Fourier method.
    2. Pseudo-spectral method by Fornberg and Whitham.

- Spectral Methods
    1. Fourier spectral method
    2. Sine spectral method
    3. Cosine spectral method
    4. Hermite spectral method
    5. LaguerreFourierHermite spectral method [25]

2.4.2 Analytical solutions

IntroductionThe experimental realization of dark solitons bright solitons and recently gap solitons in Bose-Einstein condensates has stimulated intense interest in their properties particularly their formation and propagation.Due to the non-linearity arising from the inter-atomic interactions and due to the presence of a confining potential in the NLSE or Gross-Pitaevskii equation(GPE), Sasa-Satsuma equation (SSE), and Hirota equation(HE) that describes the evolution of the solitons, these studies were performed either by solving the corresponding equations numerically or by using perturbative methods.

Interestingly enough, some analytical solutions (or exact solitonic solutions) were recently found for these equations in one dimension, but with time and space dependent inter-atomic interactions and trapping potential strengths. Obtaining such exact solutions allows for testing the validity of these NLS equations at high densities, obtain the long-time evolution of the soliton where numerical techniques may fail, and helps to understand soliton formation and propagation.

Due to the advent of high power optical soliton, higher-order terms become very important in describing the soliton propagation accurately through the fiber. The role of the higher-order terms in the stability and dynamics of the solitons thus needs to be understood. To that end, we consider here the most general form of the NLSE with cubic non-linearity, complex potential, and derivative terms. There are many different approaches to solve the nonlinear partial differential equations, some of which are developed very recently. A few of these approaches are,[20]

1. Similarity transformation method
2. Hirota bilinear method
3. The generalized and improved (G'/G)-expansion method
4. The mapping method and
5. Darboux Transformation (DT) method
6. Inverse scattering transformation method
7. Symmetry reduction method
8. The Painlev analysis method
9. the Bäcklund transformation method
10. the separated variable method etc.

We will solve this general equation with the DT method.This so-called Darboux transformation method is used to obtain analytical solutions or exact solutions. Now we will explain few of this methods in brief.

Darboux Transformation (DT) method:

The DT is one of the most powerful methods for constructing multisoliton and localized coherent structure solutions of nonlinear integrable equations in both 1+1 and 2+1 dimensions, such as the Kortewegde Vries (KdV) equation, the Kadomtsev-Petviashvili equation, the Davey-Stewartson equation, the Nizhnik-Novikov-Veselov equation, and other integrable systems . These DTs are based on the existence of Lax pairs which encode the nonlinear equations under consideration in the form of their compatibility conditions. The DT is a purely algebraic method which can be iterated step by step to generate infinitely many solutions (nontrivial) with the simplest (trivial) seed solution of the Lax pair.In the next chapter, we will describe Darboux transformation elaborately.

Hirota bilinear method

In 1971 Hirota introduced a new direct method for constructing multisoliton

solutions to integrable nonlinear evolution equations. The idea was to make a transformation into new variables, so that in these new variables multisoliton solutions appear in a particularly simple form. The method turned out to be very effective and was quickly shown to give N-soliton solutions to the Kortewegde Vries (KdV), modified Kortewegde Vries (mKdV), sine-Gordon (sG) and nonlinear Schrödinger (nlS) equations. It is also useful in constructing their Backlund transformations. Later it was observed that the new dependent variables have very good properties and this has become a starting point for further developments. In this talk our aim is to describe how multisoliton solutions can be constructed using Hirotas method. Multisoliton solutions can, of course, be derived by many other methods, e.g., by the inverse scattering transform (IST) and various dressing methods. The advantage of Hirotas method over the others is that it is algebraic rather than analytic. The IST method is more powerful (it can handle general initial conditions) and at the same time more complicated. Accordingly, if one just wants to find soliton solutions, Hirotas method is the fastest in producing results.

Single soliton solutions can be found easily by using the travelling wave ansatz f $= f(x_i - vt)$, but what about multisoliton solutions? First of all it must be stated that explicit N-soliton solutions can only be found for integrable equations. If one is only interested in finding multisoliton solutions the best tool is Hirota's bilinear method, although many other methods can also be used.

Similarity Transformation Method

We use a similarity transformation connecting some families of Nonlinear Schrödinger equations with time-varying coefficients with the autonomous cubic nonlinear Schrödinger equation. This transformation allows one to apply all known results for that equation to the non-autonomous case with the additional dynamics introduced by the transformation itself. In particular, using stationary solutions of the autonomous nonlinear Schrödinger equation we can construct exact breathing solutions to multidimensional non-autonomous nonlinear Schrödinger equations. An application is given in which we explicitly construct time dependent coefficients leading to solutions displaying weak collapse in three-dimensional scenarios. Our results can find physical applicability in mean field models of Bose-Einstein condensates and in the field of dispersion-managed optical systems.

the Backlund transformation method
These are formulas that allow one to transform one seed solution of an integrable equation into another solution that is usually more interesting. For example, taking the zero solution of the KdV equation as the seed, a Backlund transformation generates the soliton solution. Every Backlund transformation of solutions goes hand-in-hand with a transformation of the auxiliary functions simultaneously solving the equations of the Lax pair; the latter is called a Darboux transformation. Darboux transformations can be iterated, thereby generating more and more complicated exact solutions. A current area of intense interest arises from the use of Backlund transformations on the focusing nonlinear Schrödinger equation, taking the seed solution to be independent of x and nonzero (and hence representing a at pedestal that turns out to be dynamically unstable). The application of Backlund transformations to this seed, and if necessary taking certain limits in the parameters afterwards, leads to solutions that have been proposed as models of infamous rogue waves. The most famous such solution is called the Peregrine breather.

Inverse scattering method
Inverse-scattering methods for equations in higher dimensions
Several integrable equations in higher space dimensions have been discovered. These include the Kadomtsev-Petviashvili (KP) equation and the Davey-Stewartson (DS) equations. Another somewhat artificial example is the Novikov-Veselov equation (it is artificial compared with the others because it was derived directly from a Lax formulation rather than from a physical problem after which the Lax formulation was fortunately discovered; however the spectral problem that underlies the Novikov-Veselov equation is itself one of physical interest, namely the two-dimensional Schrödinger equation). The spectral parameter in these multidimensional problems enters in differently than it does in the case of integrable PDE in 1+1 dimensions. However, these equations also have other interesting mathematical properties; for example the algebrogeometric solutions of the KP equation are particularly transparent.
Inverse-scattering methods for multi-dimensional dispersionless equations
If we neglect the dispersion term in the KdV equation $u_t + uu_x + u_{xxx} = 0$, we get the inviscid Burgers or Hopf equation $u_t + uu_x = 0$, which can be solved by the method of characteristics. If we do the same with the KP

equation $(u_t + uu_x + u_{xxx})_x = u_{yy}$, we get the dispersionless KP (dKP) equation $(u_t + uu_x)_x = u_{yy}$ which does not yield to such a simple method of analysis. However, Manakov and Santini have developed a theory of the dKP and other such equations, which has several parallels with the inverse-scattering method. Their method replaces the linear equations of a Lax pair with commuting vector fields (nonlinear ODE) and replaces the linear Riemann-Hilbert problem of inverse-scattering with a certain nonlinear Riemann-Hilbert problem. Their method has many analytical gaps that remain to be filled, but it has yielded some interesting special results already.

Chapter 3

Darboux Transformation and Lax Pairs

3.1 Introduction

The Darboux transformation, originating from the work of Darboux in 1882 on the Sturm-Liouville equation, is a powerful method for constructing solutions for integrable systems. The theory is presented in several monographs and review papers. In the literature, various approaches have been proposed to find a Darboux transformation for a given equation, for instance, the operator factorization method, the gauge transformation method, and the loop group transformation.
It is remarked that the Darboux transformation is very efficient for construction of soliton solutions. Indeed, through iterations, one is often led to compact representations in terms of special determinants such as the Wronskian or Grammian for N-soliton solutions. Such N-soliton solutions are appealing both from the theoretical viewpoint and from the practical application viewpoint.[6]
In addition to the soliton solutions, rational solutions are interesting and the Darboux transformation may be adopted for this purpose. For the celebrated Korteweg-de Vries (KdV) equation, Matveev introduces the so-called generalized Darboux transformation and the positon solutions are calculated. Recently, rogue waves, appearing in oceans, have been studied and applied extensively in other fields such as Bose-Einstein condensates, optics, and superfluids. The very first model for rogue waves was the focusing nonlinear Schrödinger (NLS) equation,

$$iq_t + 12q_{xx} + |q|^2 q = 0,$$

which has been an important integrable equation. The simplest rogue wave solutions were calculated by Akhmediev and coworkers, and the construction

of higher order analogs is one of the challenges. In this regard, as pointed out by Dubard et al., the solutions obtained by Eleonskii, Krichever, and Kulagin represent a class of multi-rogue-wave solutions. It is remarked that the Darboux transformation method proposed is very specific and technical, so it may not be easy to apply to other models.
Darboux transformation use lax pair to find the solution of the nonlinear equations.We will The aim of this paper is to propose a simple method for construction multi-rogue-wave solutions. The main tool is the generalized Darboux transformation [6].

3.2 Darboux Transformation

Using the Darboux transformation, explicit multisoliton solutions (one to three soliton solutions) are obtained from a trivial initial solution [**?**].
The Darboux transformation is one of the most important tools for generating the solution of an integrable cases of quantum mechanical equations. Originally developed as a mathematical transformation that relates solutions of ordinary differential equations to each other. in the context of quantum mechanics the Darboux transformation was first found to be applicable to the linear stationary Schrödinger(NLSE) equation Ever since then, Darboux transformations have been constructed for a variety of linear and nonlinear equations, e.g. the nonlinear Schrödinger equation ,the Korteweg-de-Vries equation ,Hirota equation(HE) , sasa Satsuma equation(SSE) , higher order nonlinear Schrödinger equation HNLSE and many more.

3.2.1 History

The explicit study of surfaces of constant negative total curvature goes back to the work of Minding in 1838. Thus, in that year, Minding's theorem established the important result that these surfaces are isometric, that is, points on two such surfaces can be placed in one-to-one correspondence in a way that the metric is preserved. Beltrami subsequently gave the term pseudo-spherical to these surfaces and made important connections with Lobachevski's non-Euclidean geometry.

It was Bour, in 1862, who seems to have first set down what is now termed

the sine-Gordon equation arising out of the compatibility conditions for the Gauss equations for pseudo-spherical surfaces expressed in asymptotic coordinates.[18]

In 1879, Bianchi in his habilitation thesis presented, in mathematical terms, a geometric construction for pseudo-spherical surfaces. This result was extended by Backlund in 1883 to incorporate a key parameter which allows the iterative construction of such pseudo-spherical surfaces. The Backlund transformation was subsequently shown by Bianchi, in 1885, to be associated with an elegant invariance of the sine-Gordon equation. This invariance has become known as the Backlund transformation for the sine-Gordon equation. It includes an earlier parameter-independent result of Darboux.The Backlund transformation has important applications in soliton theory. Indeed,it appears that the property of invariance under Backlund and associated Darboux transformations as originated in,is enjoyed by all soliton equations. The contribution of Bianchi and Darboux to the geometry of surfaces and, in particular, the role of Backlund transformations preserving certain geometric properties have been discussed by Chern and Sym et al. It is with Backlund and Darboux transformations, their geometric origins and their application in modern soliton theory that we shall be concerned in the present monograph.[18]
One of the basic techniques for solving differential equations, that of reducing them to equations whose solutions are known, has a long history. It probably began with the work of Kummer and Liouville, who solved the problem of reducing a second order differential equation with variable coefficients to an equation of previously specified form. The transformation which they used, which from that time has been known as the Kummer-Liouville transformation, is the most general transformation which preserves the linearity and order of the equation. [3] A modern, systematic discussion of the results in this area can be found in Ref.[10]

3.2.2 Definition of Darboux Transformation

Analytical Definition

Darboux transformation is one of the methods used in solving nonlinear evolution equation. Basically, the Darboux transformation is a linear algebra formulation of the solutions of the Zakharov-Shabat system of equations associated with the nonlinear evolution equation. The evolution of monochromatic electromagnetic wave in a nonlinear-dispersive optical medium is considered in this case[**?**].

Mathematical Definition

Consider a Sturm-Liouville equation
$-\psi_{xx}(x) + u(x)\psi = \lambda\psi(x)$,

with known fixed eigensolution ψ_1 and eigenvalue λ_1. .....(1)

The Darboux transformation on ψ is defined as:

$\psi_[1] = (\frac{d}{dx} - \frac{\psi_{1x}}{\psi_1})\psi$ .....(2)

3.2.3 Theorem:

The function $\psi_[1]$ satisfies
$-\psi[1]_{xx} + u[1]\psi[1] = \lambda\psi[1]$, ...(3)

with

$u[1] = u - 2(\frac{\psi_{1x}}{\psi_1})_x$.....(4) ..

Eq.(1) is said to be covariant under the DT.[14]

Our objective in this paper is to use the Darboux transformation method to get exact solitonic solutions for higher order nonlinear Schrödinger equation HNLSE with complex potentials, and time-dependent coeficients. cubic

and quintic nonlinearities together .

3.2.4 DT for Linear Ordinary Differential Equations

Schrödinger Equation With a Harmonic Potential

Consider the equation
$-\psi_{xx} + x^2\psi = \lambda\psi$, ...(5)

where, $u = x^2$
and ψ is eigenfunction and λ is eigenvalue.
To apply darboux transformation(DT) on the eigenfunction ψ we need a seed solution ψ_1 and an eigenvalue λ. If $\psi_1 = e^{-\frac{x^2}{2}}$ and $\lambda = -1$ then according to equation (2), applying DT on ψ_1 we get

$\psi[1] = (\frac{d}{dx} - \frac{\psi_{1x}}{\psi_1})\psi$

$\Rightarrow \psi[1] = (\frac{d}{dx} - \frac{\frac{d}{dx}e^{-\frac{x^2}{2}}}{e^{-\frac{x^2}{2}}})\psi$

$\Rightarrow \psi[1] = (\frac{d}{dx} - x)\psi$ .....(6a)

And from equation (4),

$u[1] = u - 2\frac{d}{dx}(\frac{\frac{d}{dx}e^{-\frac{x^2}{2}}}{e^{-\frac{x^2}{2}}})$

$\Rightarrow u[1] = x^2 - 2$.....(6b)

which satisfy the equation

$-\psi[1]_{xx} + x^2\psi[1] = (\lambda + 2)\psi$.....(7)

We can prove it,

$L.H.S = -\psi[1]_{xx} + x^2\psi[1]$

$= -\frac{d}{dx}\frac{d}{dx}(\frac{d}{dx} - x)\psi + x^2(\frac{d}{dx} - x)\psi$

$= -\frac{d}{dx}(\frac{d^2\psi}{dx^2} + \psi + x\frac{d\psi}{dx}) + x^2(\frac{d}{dx} - x\psi)$

$= -\frac{d^3\psi}{dx^3} + \frac{d\psi}{dx} + \frac{d\psi}{dx} + x\frac{d^2\psi}{dx^2} + x^2\frac{d\psi}{dx} - x^3\psi$

$= -\frac{d^3\psi}{dx^3} + 2\frac{d\psi}{dx} + x^3\psi - x\lambda\psi + x^2\frac{d\psi}{dx} - x^3\psi$

$= -\frac{d}{dx}(x^2\psi - \lambda\psi) + 2\frac{d\psi}{dx} - x\lambda\psi + x^2\frac{d\psi}{dx}$

$= -\frac{d}{dx}(x^2\psi - \lambda\psi) + 2\frac{d\psi}{dx} - x\lambda\psi + x^2\frac{d\psi}{dx}$

$= -2(\frac{d\psi}{dx} - x)\psi + \lambda(\frac{d\psi}{dx} - x)\psi$

$= (\lambda + 2)(\frac{d\psi}{dx} - x)\psi$

$= (\lambda + 2)\psi[1]$

=R.H.S Here, from equation (5)

$= -\frac{d^3\psi}{dx^3} = \frac{d}{dx}(\frac{d^2\psi}{dx^2}) = -\frac{d}{dx}(x^2\psi - \lambda\psi)$

For example,

If $\psi = e^(-x^2/2)$ and $\lambda = 1$, then applying darboux transformation on ψ we get

$\psi[1] = (\frac{d}{dx} - x)e^(-x^2/2)$
$= -2xe^{-x^2/2}$ .....(8)

Now, with eigenvalue $\lambda + 2 = 3$.

Applying the DT on $\psi[1]$, we get

$\psi[2] = (\frac{d}{dx} - x)\psi[1]$
$= (\frac{d}{dx} - x)-2xe^{-x^2/2}$

$= \frac{d}{dx} -2xe^{-x^2/2} - x-2xe^{-x^2/2}$
$= -2x^2e^{-x^2/2} - 2e^{-x^2/2} + 2x^2e^{-x^2/2}$
$= -(4x^2 - 2)e^{x^2/2}$.....(9)

Now,with eigenvalue $\lambda + 2 = 5$ we can apply DT on $\psi[2]$

Applying the DT on $\psi[n]$ n-times, we get

$\psi[n] = -H_n e^(x^2/2)$.....(10)
with eigenvalue $\lambda + 2 = 2n + 1$, where
$H_n = (-1)^n e^(x^2/2)(d/dx - x)^n e^(- x^2/2)$
is the Hermite polynomial of order n. .....(11)

3.2.5 How to use Darboux Transformation?

The method of Darboux transformation can be used to obtain exact solutions of the Gross-Pitaevskii equation with quadratic expulsive potential and scattering length that is growing exponentially with time.The Darboux transformation, or analogously Backlund or dressing transformation,applies only to systems of linear differential equations and cannot be applied directly to nonlinear differential equations. To be able to apply the Darboux transformation to a certain nonlinear differential equation, one finds a linear system of equations that is equivalent to a nonlinear differential equation. The relation between the linear system and the nonlinear differential equation is established through a consistency condition satisfied by the linear system.The Darboux transformation is then applied to the linear system resulting in transforming the equivalent nonlinear equation as well. The linear system is usually represented in terms of a pair of matrices called the Lax pair which must satisfy a consistency condition that is equivalent to the differential equation at hand. The difficulty is usually in finding this Lax pair.[12]

It is known for some nonlinear differential equations such as the Kortweg-de Vries (KdV) equation, the sine-Gordon equation and the nonlinear Schrödinger equation. In addition to the Lax pair, one also needs to know an exact solution of the nonlinear differential equation. This exact solution is then used as a seed for the Darboux transformation to generate other exact solutions.

We present a method of obtaining these two essential ingredients for performing the Darboux transformation, namely the Lax pair and the seed solution. In addition to some known Lax pairs for the Gross-Pitaevskii equation, our method successfully produces ones. In addition to the interest in constructing new nontrivial solitonic solutions from simpler ones, there is also interest in obtaining generalized Lax pairs of modified Gross-Pitaevskii equations concerning integrable and chaotic behaviour.[13, 27]

3.2.6 Application of Darboux Transformation

The application of this transformation to the Schrödinger equation involves some special features can be found in Ref.[17]. The most complete study of Schrödinger equations reduced to the hyper geometric equation can be found.Transformations which raise the order of an equation have been studied by Moutard and Imshenetski. The first systematic study of such transformations was undertaken by Darboux, and so these transformations have been named after him. They were subsequently rediscovered many times. For example, the factorization method, which became well known after the studies by Schrödinger and has been studied systematically by Infeld and Hull is essentially another formulation of the Darboux transformation[2]. Super symmetric quantum mechanics, first introduced by Witten, uses transformations which intertwine the components of the super-Hamiltonian, and these are Darboux transformations of the original Schrödinger equation.Mention should also be made of the Abraham-Moses-Pursey integral transformations, which represent an application of the methods of inverse scattering theory for generating exactly solvable potentials of the one-dimensional stationary Schrödinger equation. In the most important and almost unique case where it is possible to obtain a closed-form solution of the Gel'fand-Levitan-Marchenko equations the case of degenerate kernel, these transformations are equivalent to a special case of the generalized Darboux transformations. Owing to this fact, a number of the results obtained by the inverse-scattering method can be reproduced by using Darboux transformations. This approach has certain advantages arising from the differential nature of the transformation. Especially attractive is the idea of using qualitative methods of controlling spectra, developed on the basis of inverse-scattering methods, together with Darboux transformations, which allow solutions of the transformed Schrödinger equation to be obtained fairly simply. The method of

Darboux transformations is undergoing active development at the present time. A hyperbolic equation with two independent variables is studied and a second-order Darboux transformation for the one and two dimensional stationary Schrödinger equations is studied in detail the Darboux transformations associated with the scattering problem in quantum mechanics. References study the development of this method for the one-dimensional non-stationary Schrödinger equation, which was first introduced in connection with the solution of the Kadomtsev-Petviashvili equation [2].

3.2.7 Darboux Transformation of the Non-stationary Schrödinger Equation

Let us consider the Schrödinger equation for a particle with potential energy $-2V_0(x,t)$:

$$\begin{aligned} (i\partial_t - h_0)\psi(x,t) &= 0 \\ h_0 = -{\partial_x}^2 - V_0(x,t) & \\ \partial = \frac{\partial}{\partial t}, \quad {\partial_x}^2 = \partial_x.\partial_x, \quad x \in [a,b] & \text{.....(1)} \end{aligned}$$

We shall say that the operator $\hat{L}$ defined on solutions of Eq.(1) is a transformation operator for this equation if it satisfies the operator equation

$\hat{L}(i\partial_t - h_0) = (i\partial_t - h_1)\hat{L}$, ... (2)
where $h_1 = -{\partial_x}^2 - V_1(x,t)$ is the transformed Hamiltonian.

The operator $\hat{L}$ defined by (2) transforms every solution $\psi(x,t)$ of Eq.(1) into a solution $\phi(x,t) = \hat{L}\psi(x,t)$ of the transformed Schrödinger equation:

$(i\partial_t - h_1)\phi(x,t) = 0, \quad s \in [a,b]$.....(3)
We use T_0 and T_1 to denote the linear spaces of solutions of Eqs. (1) and (3), respectively.If the conjugation operator [denoted by the symbol $^+$], which possesses the property $(AB)^+ = B^+A^+$ and under which the Schrödinger operators $i\partial_1 - h_{0,1}$ are self-adjoint, $(i\partial_t - h_1)^+ = i\partial_1 - h_{0,1}$ is defined for the operators acting in the spaces T_0 and T_1;

then it follows from (2) that

$(i\partial_t - h_0)\hat{L}^+ = \hat{L}^+(i\partial_t - h_1)$ ..... (4)

In this case the operator $\hat{L}^+$ effects a transformation in the reverse direction, i.e., from solutions of (3) to solutions of (1).The product of operators $\hat{L}^+\hat{L}$ will obviously transform the solutions of(1) into other solutions of (1) and will therefore be a symmetry operator of this equation.In complete analogy, $\hat{L}\hat{L}^+$ will be a symmetry operator of Eq. (3). If $\hat{L}$ is a differential operator in the variables x and t, then by solving (2) in the space T_0 we can replace $i\partial_t$ in it by h_0 .Therefore, the operator $\hat{L}$ can safely be assumed to be a differential operator in the variable x whose coefficients are functions of the variables x and t.

A differential operator of order N in the variable x with coefficients depending on x and t which satisfies Eq.(2) will be called an Nth-order Darboux transformation operator.For N=1 we shall simply call it a Darboux transformation operator.
The class of operators defined only by Eq. (2) is larger than the class of operators transforming one solution of the KadomtsevPetviashvili equation

$\partial_x(V_t + 6VV_x + V_{xxx}) = 3V_{yy}$ .....(5)

into another solution, which were studied earlier in Ref.[11] and [15]. This fact can be explained as follows. Equation (5) is the compatibility condition for two linear differential equations the Lax representation; see, for example, Ref.[15],

$$B^{(}0)\psi(x,y,t) = 0, \quad B^{(}0) = i\partial_y + {\partial_x}^2 + V^{(}0),$$
$$A^{(}0)\psi(x,y,t) = 0, \quad A^{(}0) = \partial_t + 4{\partial_x}^3 + 6V^{(}0)\partial_x + 3V^{(}0) + 3iw^{(}0), \quad \text{.....(6)}$$

imposed on the function $V^{(}0) = V^{(0)}(x,y,t)$. Equation (6) is a non-stationary Schrödinger equation in which the variable y plays the role of the time. If now, following Zakharov and Shabat, we find the operator L, which is involved simultaneously in two coupled equations,

$$LB^{(}0) = B^{(}1)L,$$
$$LA^{(}0) = A^{(}1)L, \quad \text{.....(7)}$$

where $B^{(}1)$ and $A^{(}1)$ differ from $B^{(}0)$ and $A^{(}0)$ only by the replacement of $V^{(}0)$ by $V^{(}1)$ and $w^{(}0)$ by $w^{(}1)$, then the system of equations

$B^{(}1)\phi = 0, \quad A^{(}1)\psi = 0, \phi = L\psi$
will be compatible and the function $V = V^{(}1)(x, y, t)$ will be a solution of (5). From this it is clear that if we reject the condition (7), we enlarge the class of operators L.

3.3 Lax Pair

3.3.1 Definition

A pair of linear operators L and A associated with a given partial differential equation which can be used to solve the equation. However, it turns out to be very difficult to find the L and A corresponding to a given equation, so it is actually simpler to postulate a given L and A and determine to which partial differential equation they correspond.In mathematics, in the theory of integrable systems, a Lax pair is a pair of time-dependent matrices or operators that satisfy a corresponding differential equation, called the Lax equation. Lax pairs were introduced by Peter Lax to discuss solitons in continuous media. The inverse scattering transform makes use of the Lax equations to solve such systems.
A Lax pair is a pair of matrices or operators L(t), P(t) dependent on time and acting on a fixed Hilbert space, and satisfying Lax's equation:

$\frac{dL}{dt} = [P, L]$
where [P,L]=PL-LP is the commutator. Often, as in the example below, P depends on L in a prescribed way, so this is a nonlinear equation for L as a function of t.

3.3.2 Lax Equation

Lax equation is used in integrable systems; namely some systems are equivalent to the Lax equation. Lax equation is a linear ordinary differential equation of the form

$\frac{dL}{dt} = [M, L]$
for $n \times n$-matrix-valued function L=L(t)L = L(t), where MM is also a $n \times n$ matrix. The pair (L,M)(L,M) is also called a Lax pair. The Lax equation is the compatibility condition for the system

$\lambda\psi = L\psi$
$\frac{d\psi}{dt} = M\psi$, where $\psi = \psi(t)$ is a vector function which is an eigenvector for LL with eigenvalue λ. To see this make a derivative of $L\psi$ and use the Leibniz rule.

$$ML\psi = M\lambda\psi = \lambda M\psi = \lambda\frac{d\psi}{dt} = d\frac{L\psi}{dt} = \frac{dL}{dt}\psi = \frac{dL}{dt}\psi + L\frac{d\psi}{dt} = \frac{dL}{dt}\psi + LM\psi$$

Example

The Lax formalism for the KdV equation. Consider the differential operators,

$L = -6\frac{d^2}{dx^2}$,
$B = -4\frac{d^3}{dx^3} - u\frac{d}{dx} - \frac{1}{2}u_x$
L is symmetric (formally self-adjoint with respect to the L^2 inner product), and B is skew-symmetric (formally skew-adjoint). Then it was noted by Peter Lax in 1968 that the operator equation.
$\frac{dL}{dt} + [L, B] = 0$
.....(1)

The equation (1) can be considered as the consistency condition of the following linear system for an auxiliary field ψ which is equivalent to the following KdV equation
$u_t + uu_x + u_{xxx} = 0$

Now, applying the operator equation on a function $\psi(x, t)$ we get,
$[\frac{dL}{dt} + [L, B]]\psi(x, t) = 0$
$\Rightarrow \frac{dL}{dt}\psi + [L, B]\psi = 0$
$\Rightarrow \frac{dL}{dt}\psi + (LB - BL)\psi = 0$ .....(2)

Here,

$\frac{d}{dt} = D_t$

$\frac{d}{dx} = D_x$
Now,

$\frac{dL}{dt}\psi = \frac{d}{dt}(L\psi) - L\frac{d}{dt}\psi$

$= \frac{d}{dt}[(-6\frac{d^2}{dx^2}\psi) - 6\frac{d^2}{dx^2}\frac{d}{dt}\psi]$

$= D_t u\psi$ [Eigenvalues of L are independent of t]

$= u_t\psi$

$(LB - BL)\psi = (LB)\psi - (BL)\psi$

$= (-6{D_x}^2 - u)(-4{D_x}^3 - uD_x - \frac{u_x}{2})\psi - (-4{D_x}^3 - uD_x - \frac{u_x}{2})(-6D_x - u)\psi$

$= (6{D_x}^2 + u)(4{D_x}^3 + uD_x + \frac{u_x}{2})\psi - (4{D_x}^3 + uD_x + \frac{u_x}{2})(6{D_x}^2 + u)$

$= 24{D_x}^5\psi + 6{D_x}^2(uD_x\psi) + 3{D_x}^2(u_x\psi) + 4u{D_x}^3\psi + + \frac{1}{2}u(D_xu)\psi + 24{D_x}^5\psi - 4{D_X}^3U\psi - 6u{D_x}^3\psi - u^2D_x\psi - 3u_x{D_x}^2\psi - u\frac{u_x}{2}\psi$

$= 6{D_x}^2(uD_x)\psi + 3{D_x}^2(u_x\psi) + 4u{D_x}^3\psi + u^2(D_x\psi) + \frac{1}{2}u(D_xu)\psi - 4{D_x}^3u\psi - 6u{D_x}^3\psi - u^2D_x\psi - 3u_x{D_x}^2\psi - u\frac{u_x}{2}\psi$

Now,
Here,

$6{D_x}^2(uD_x)\psi$

$= 6{D_x}^2(uD_x\psi)$

$= 6D_x[D_x(uD_x\psi)]$

$= 6D_x[(D_xu)D_x\psi + u({D_x}^2\psi)]$

$= 6[D_x(D_xu)D_x\psi + D_x(u{D_x}^2\psi)]$

$= 6[D_x(D_xu)\psi_x + D_x(u{D_x}^2)\psi]$

$= 6[({D_x}^2 u)\psi_x + D_x(D_x u)\psi_x + {D_x}^2(D_x u)\psi + u({D_x}^3\psi)]$

$= 6[({D_x}^2 u)\psi + (D_x u)D_x\psi_x + (D_x u){D_x}^2\psi + ({D_x}^3\psi)u]$

$= 6(u_{xx}\psi_x + u_x\psi_{xx} + u_x\psi_{xx} + \psi_{xxx}u)$

$= 6u_{xx}\psi_x + 12u_x\psi_{xx} + 6\psi_{xxx}u$

and,

${3D_x}^2(u_x\psi)$

$= 3D_x[D_x(u_x\psi)]$

$= 3D_x[(D_x u_x)\psi + (u_x D_x)\psi]$

$= 3[D_x(D_x u_x)\psi + D_x(u_x D_x)]$

$= 3[D_x(D_x u_x)\psi + ({D_x}^2 u_x)\psi + D_x(D_x u_x)\psi + U_x({D_x}^2\psi)]$

$= 3(D_x u_x)D_x\psi + 3({D_x}^2 u_x)\psi + 3(D_x u_x)D_x\psi + 3u_x({D_x}^2\psi)$

$= 6(D_x u_x)D_x\psi + 3({D_x}^2 u_x)\psi + 3u_x({D_x}^2\psi)$

$= 6u_{xx}\psi_x + 3u_{xxx}\psi + 3u_x\psi_{xx}$

and,

${4D_x}^3 u\psi$

$= 4{D_x}^2[(D_x u)\psi + (uD_x)\psi]$

$= 4D_x[D_x(D_x u)\psi + D_x(uD_x)\psi]$

$= 4D_x[D_x(D_x u)\psi + ({D_x}^2 u)\psi + D_x(D_x u)\psi + ({D_x}^2\psi)u]$

$= 4D_x[2D_x(D_x u)\psi + ({D_x}^2 u)\psi + D_x(D_x u)\psi + ({D_x}^2\psi)u]$

$= 4D_x[2(D_xu)\psi_x + (D_x{}^2u)\psi + D_x(D_x{}^2\psi)u]$

$= 4[2D_x(D_xu)\psi_x + D_x(D_x{}^2u)\psi + D_x(D_x{}^2\psi)u]$

$= 4[2(D_xu)D_x\psi_x+2(D_x{}^2u)\psi_x+(D_xu)D_x{}^2\psi+(D_x{}^3u)\psi+(D_x\psi)D_x{}^2+(D_x{}^3\psi)u]$

$= 4[3(D_xu)D_x{}^2\psi + 3(D_x{}^2u)\psi_x + (D_x{}^3u)\psi]$

$= 12u_x\psi_{xx} + 12u_{xx}\psi_x + 4u_{xxx}\psi + 4u\psi_{xxx}$

and $u(D_xu\psi) = u[(D_xu) - uD_x]\psi = uu_x\psi - u^2\psi_x$

So, $(LB-BL)\psi = 6u_{xx}\psi_x+12u_x\psi_{xx}+6\psi_{xxx}u+6u_{xx}\psi_x+3u_{xxx}\psi+3u_x\psi_{xx}+4u\psi_{xxx} + u^2\psi_x - 4u\psi_{xxx} + u\frac{u_x}{2}\psi - 12u_x\psi_{xx} - 12u_x\psi_{xx} - 12u_{xx}\psi_x4u\psi_{xxx} - uu_x\psi - u^2\psi_x - 3u_x\psi_{xx} - u\frac{u_x}{2}\psi$

$= 12\psi_{xx}\psi_x + 3u_{xxx}\psi + 4u\psi_{xxx} - 12u_{xx}\psi_x - 4u_{xxx}\psi - 4u\psi_{xxx} - uu_x\psi$

$= -u_{xxx}\psi - uu_x\psi$

Now, from equation (2) we get,

$\frac{dL}{dt}\psi + (LB - BL)\psi = 0$

$\Rightarrow -u_t\psi - u_{xxx\psi - uu_x\psi = 0}$

So, $(u_t + uu_x + u_{xxx})\psi(x,t) = 0$

This operator equation of this form is today called a Lax equation.
Another way to derive Lax's form of the KDV equation is as the compatibility condition of two linear problems, where λ is a fixed parameter.

$L\phi = \lambda\phi$ and $\phi_t = B\phi$

$\phi = \phi(x,t)$

Now,
$\frac{d}{dt}(L\phi) = \frac{dL}{dt}\phi + L\frac{d\phi}{dt}$

$= \frac{dL}{dt}\phi + L\phi_t$

$\frac{d}{dt}(L\phi) = \frac{dL}{dt}\phi + LB\phi$.....(3)

By using $L\phi = \lambda\phi$

$\frac{d}{dt}(L\phi) = \frac{d}{dt}(\lambda\phi)$

$= \lambda\phi_t$

$= \lambda B\phi = B(\lambda\phi) = Bl\phi$

So, $\frac{d}{dt}(L\phi) = BL\phi$

which again gives equation (1).

3.3.3 Isospectral Property

It can then be shown that the eigenvalues and more generally the spectrum of L are independent of t. The matrices/operators L are said to be isospectral as t varies.

The core observation is that the matrices L(t) are all similar by virtue of

$L(t) = U(t,s)L(s)U(t,s)^{-1}$

where U(t,s) is the solution of the Cauchy problem

$\frac{d}{dt}U(t,s) = P(t)U(t,s), \qquad U(s,s) = I,$

where I denotes the identity matrix. Note that if L(t) is self-adjoint and P(t) is skew-adjoint, then U(t,s) will be unitary.

In other words, to solve the eigenvalue problem $L\psi = \lambda\psi$ at time t, it is possible to solve the same problem at time 0 where L is generally known better, and to propagate the solution with the following formulas:

$\lambda(t) = \lambda(0)$ (no change in spectrum)

$\frac{\partial\psi}{\partial t} = P\psi$

3.4 The Darboux transformation and the Lax pair

The Lax pair is the most important ingredient in solving a NLSE using the DT. Here we summarize the basic equations only. In order to derive the Lax pair we start by writing the following two equations:
$\psi_x = U_0\psi + U_1\psi\Lambda,$

$\psi t = V_0\psi + V_1\psi\Lambda + V_2\psi\Lambda^2 + V_3\psi\Lambda^3$

Now,

$\psi_{xt} = U_0 t\psi + U_0\psi_t + U_{1t}\psi\Lambda + U_1\psi_t\Lambda$.....(1)

and,
$\psi_{tx} = V_{0x}\psi + v_0\psi_x + V_{1x}\psi\Lambda + V_1\psi_x\Lambda + V_{2x}\psi\Lambda^2 + V_2\psi_x\Lambda^2 + V_{3x}\psi\Lambda^3 + V_3\psi_x\Lambda^3$.....(2)

From (1) $\Rightarrow$

$\psi_{xt} = U_0 t\psi + U_0(V_0\psi + V_1\psi\Lambda + V_2\psi\Lambda^2 + V_3\psi\Lambda^3) + U_{1t}\psi\Lambda + U_1(V_0\psi + V_1\psi\Lambda + V_2\psi\Lambda^2 + V_3\psi\Lambda^3)\Lambda$

$\Rightarrow \psi_{xt} = U_0 t\psi + U_0V_0\psi + U_0V_1\psi\Lambda + U_0V_2\psi\Lambda^2 + U_0V_3\psi\Lambda^3 + U_{1t}\psi\Lambda + U_1V_0\psi\Lambda + U_1V_1\psi\Lambda^2 + U_1V_2\psi\Lambda^3 + U_1V_3\psi\Lambda^4$

$\Rightarrow \psi_{xt} = (U_{0t}\psi + U_0V_0\psi)\Lambda^0 + (U_0V_1\psi + U_1V_0\psi + U_{it}\psi)\Lambda + (U_0V_2\psi + U_1V_1\psi)\Lambda^2 + (U_0V_3\psi)\Lambda^3 + (U_{1t}V_3\psi)\Lambda^4$.....(3)

From (2) $\Rightarrow$,

$\psi_{tx} = V_{0x}\psi + V_0(U_0\psi + U_1\psi\Lambda) + V_{1x}\psi\Lambda + V_1(U_0\psi + U_1\psi\Lambda)\Lambda + V_{2x}\psi\Lambda^2 + V_2(U_0\psi + U_1\psi_0\Lambda)\Lambda^2 + V_{3x}\psi\Lambda^3 + V_3(U_0\psi + U_1\psi\Lambda)\Lambda^3$

$\Rightarrow \psi_{tx} = V_{0x}\psi + V_0U_0\psi + V_0U_1\psi\Lambda + V_{1x}\psi\Lambda^1 + V_1U_0\psi\Lambda + V_1U_1\psi\Lambda^2 + V_{2x}\psi\Lambda^2 + V_2U_0\psi\Lambda^2 + V_2U_1\psi\Lambda^3 + V_{3x}\psi\Lambda^3 + V_3U_0\psi\Lambda^3 + V_3U_1\psi\Lambda^4$

$\Rightarrow \psi_{tx} = (V_{0x}\psi + V_0U_0\psi)\Lambda^0 + (V_0U_1\psi + V_{1x}\psi + V_1U_0\psi)\Lambda + (V_1U_1\psi + V_{2x}\psi + V_2U_0\psi)\Lambda^2 + (V_2U_1\psi + V_{3x}\psi + V_3U_0\psi+)\Lambda^3 + (V_3U_1\psi)\Lambda^4$.....(4)

Comparing the coefficient of $\Lambda^0, \Lambda^1, \Lambda^2, \Lambda^3, \Lambda^4$ from equation (3) and (4) we get,

$U_{0t}\psi + U_0V_0\psi = D_{0x}\psi + V_0U_0\psi[Coefficient of \Lambda^0]$

$\Rightarrow U_{0t} - V_{0x} + [U_0, V_0] = 0$.....(5)

$U_0V_1\psi + U_1V_0\psi + U_{it}\psi = V_0U_1\psi + V_{1x}\psi + V_1U_0\psi[Coefficient of \Lambda^1]$
$\Rightarrow U_{1t} + U_0V_1\psi + [U_0, V_1] + [U_1, V_0] = 0$.....(6)

$U_0V_2\psi + U_1V_1\psi = V_1U_1\psi + V_{2x}\psi + V_2U_0\psi[Coefficient of \Lambda^2]$

$\Rightarrow V_{2x} + [V_2, U_0] + [V_1, U_1] = 0$.....(7)

$U_0V_3\psi + U_1V_2\psi = V_{3x}\psi + V_3U_0\psi + V_2U_1\psi[Coefficient of \Lambda^3]$

$\Rightarrow V_{3x} + [V_3, U_0] + [V_2, U_1] = 0$.....(8)

$U_{1t}V_3\psi = V_3U_1\psi[Coefficient of \Lambda^4]$

$[V_3, U_1] = 0$.....(9)

where [X, Y] denotes the commutator of X and Y. Equation (3) gives back

the original NLSE, which is the condition that the Lax pair must satisfy. The most important and difficult part is to find U_0 and V_0 matrices for a given NLSE. The remaining matrices are determined from U_0 and V_0 matrices.We use a systematic search approach to find U_0 and V_0 matrices [1]. It is worth mentioning that for some higher order NLSE we find U_0 and V_0 matrices but do not find the remaining matrices.Therefore, analytical solution of the given higher-order NLSE is not possible using the DT. After deriving the set of matrices for the given NLSE we use the DT to find the analytic solution. We solve (1) and (2) for the auxiliary field $\psi = \psi_0$. Using this auxiliary field we find the solution of the original NLSE, which is given by,

$U_0[1] = U_0 + [U_1, \sigma]$,.....(9)

where $U_0[1]$ is the new U_0 matrix and $\sigma = \Psi_0.\Lambda.{\Psi_0}^{-1}$. This $U_0[1]$ contains the solution of the given NLSE. The beauty of the DT lies in the fact that a trivial seed solution, for example,zero or continuous wave solution of the given NLSE, leads to a nontrivial solution.

Chapter 4

Different Solutions of Simplest Nonlinear Schrödinger Equation

The nonlinear Schrödinger equation is an example of a universal nonlinear model that describes many physical nonlinear systems.There are many types of solutions of this nonlinear Schrödinger equation.In this chapter we will find some solutions of the simplest nonlinear schrödinger equation.[25]

4.1 NLSE with Cubic Non-linearity

In this section we consider the most general NLSE with qubic non-linearity [1, 25-28]. In the presence of damping or gain and external potential, the general NLSE can be written as

$i\psi_t + a_1\psi_{xx} + a_2|\psi|^2\psi + ia_3\psi_{xxx} + ia_4|\psi|^2\psi$

$+ia_5|\psi|^2{\psi_x}^* + i\Gamma\psi + V_{ext}\psi = 0$.....(3.1)

This equation describes the evolution of the electric field envelope of an optical pulse through an optical fiber. The second term denotes the chromatic or group velocity dispersion for which the optical pulse wants to be stretched and dispersed, the third term denotes the Kerr non-linearity or self-phase modulation (arises due to intensity dependent refractive index) for which the shape and size of soliton remain same over time, and the fourth term denotes the third-order dispersion. The fifth and the sixth terms are related to the self steepening and self-frequency shift due to the Raman scattering.

So we have to know about these terms stated in the above nonlinear schrödinger equation.A brief description is given below:

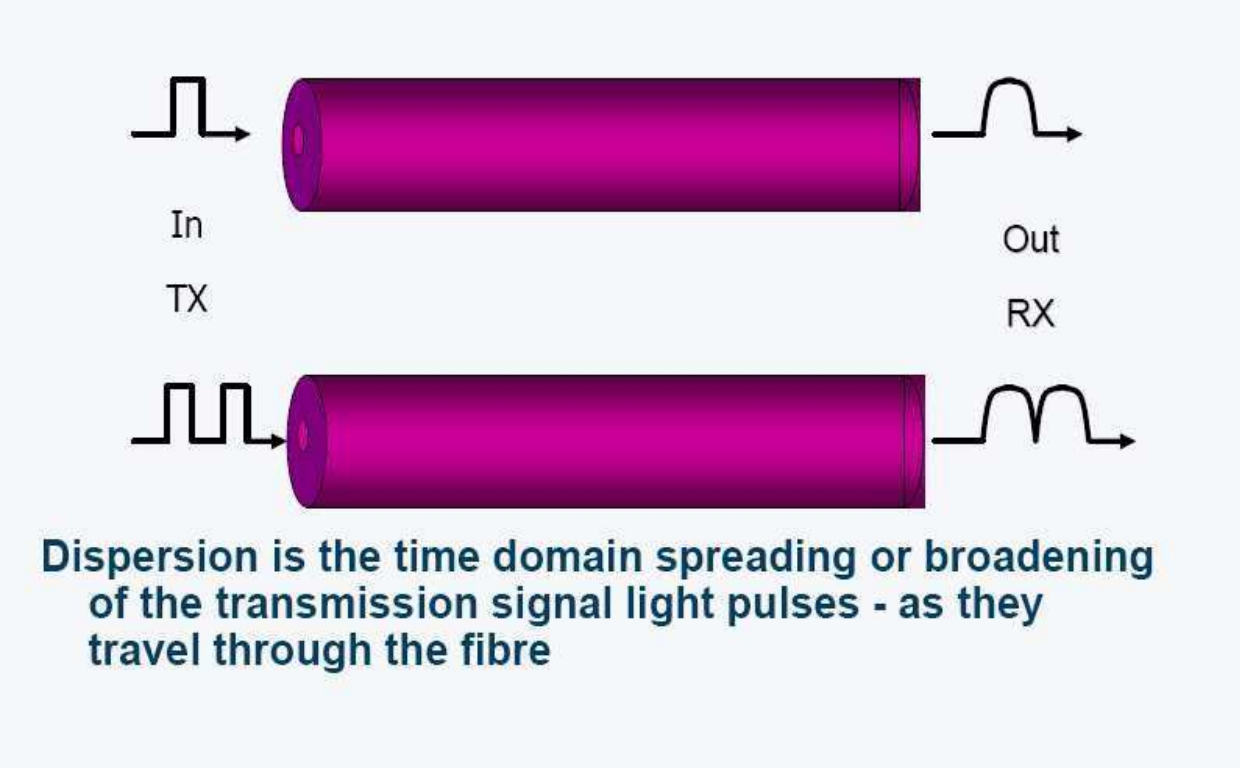

Figure 4.1: Chromatic dispersion in optical fiber.

4.1.1 Chromatic Dispersion

Chromatic dispersion is a phenomenon that is an important factor in fiber optic communications. Sir Isaac Newton observed this phenomenon when he passed sunlight through a prism and saw it diverge into a spectrum of different colors. This dispersion occurs because different colors, or light frequencies, act slightly differently as they pass through a medium such as glass. In fiber-based systems, an optical fiber, comprised of a core and cladding with differing refractive index materials, inevitably causes some wavelengths of light to travel slower or faster than others.[22]

So what effect does the chromatic dispersion has on fiber optics communication system?
The direct effect is light pulse broadening.In the figure[4.1]A perfect light

pulse starts at the beginning. After it travels the length of fiber, may it be of 100m or 2000m, the light pulse become wider than the original signal.This is caused by chromatic dispersion.Because, in the same pulse blue light travels faster and the red light travels slower. So the total width of the pulse get stretched wider.Now, look at the bottom part of the figure[4.1].Here, two perfect signals starts at the begining, but at the end, since both pulses get wider because of chromatic dispersion they get overlapped with each other.If the overlap is too big the detector will not be able to recognize these two signals and you will get an error.So, chromatic dispersion causes pulse broadening and a bit error.

4.1.2 Kerr non-linearity

The optical Kerr effect, or AC Kerr effect is the case in which the electric field is due to the light itself.When light passes through the optical fiber the corresponding electric field changes the refractive index of the core and cladding(which is basically made of silica). This causes a variation in index of refraction which is proportional to the local irradiance of the light(i.e; the light's amplitude or strength). This refractive index variation is responsible for the nonlinear optical effects of self-focusing, self-phase modulation and modulational instability, and is the basis for Kerr-lens modelocking.If the pulse has just the right shape, the Kerr effect will exactly cancel the dispersion effect, and the pulse's shape will not change over time.

4.1.3 Third-order Dispersion

Third-order Dispersion results from the frequency dependence of the Group Delay dispersion. In the expansion of the Spectral Phase versus angular frequency offset, it is related to the third-order term. In mode-locked lasers for pulse durations below roughly 30 femtoseconds, it is necessary to provide Dispersion Compensation not only for the group delay dispersion, but also for the third-order dispersion and possibly for even higher orders. In many cases, the investigation of the effect of third order dispersion requires numerical pulse propagation modeling.

4.1.4 Self-Steepening

Self-steepening of optical pulses arises due the dispersive contribution of the Kerr nonlinearity. In typical structures this response is on the order of a few femtoseconds with a fixed frequency response. In contrast, the effective Kerr nonlinearity in photonic crystal waveguides (PhCWGs) is largely determined by the geometrical parameters of the structure and is consequently tunable over a wide range.

4.1.5 Self-frequency Shift

When propagating in an optical fiber, soliton pulses are subject not only to the Kerr nonlinearity, but also to stimulated Raman scattering. For very short solitons (with durations of e.g. ¡100 fs), the optical spectrum becomes so broad that the longer-wavelength tail can experience Raman amplification at the expense of power in the shorter-wavelength tail. This causes an overall spectral shift of the soliton towards longer wavelengths, i.e., a soliton self-frequency shift. The strength of this effect depends strongly on the pulse duration, since shorter solitons exhibit a higher peak power and a broader optical spectrum. The latter is important because the Raman gain is weak for small frequency offsets. During propagation, the rate of the frequency shift often slows down, because the pulse energy is reduced and the pulse duration increased.[26]

The soliton self-frequency shift can be exploited for reaching spectral regions which are otherwise difficult to access. By adjusting the pulse energy in the fiber, it is possible to tune the output wavelength in a large range

4.2 Why do we need Non-linear Schrödinger equation?

In the previous chapters we have discussed about different fields , where we can apply nonlinear Schrödinger equation(NLSE).In this section we will be more specific about some fields and will try to be clear about it.
Firstly, NLSE is applied in optics,specificly in fiber optics. For examples, In the land-based long-distance fiber telephone network, installing submarine fiber telephone cables beneath the major oceans, developing fiber optic

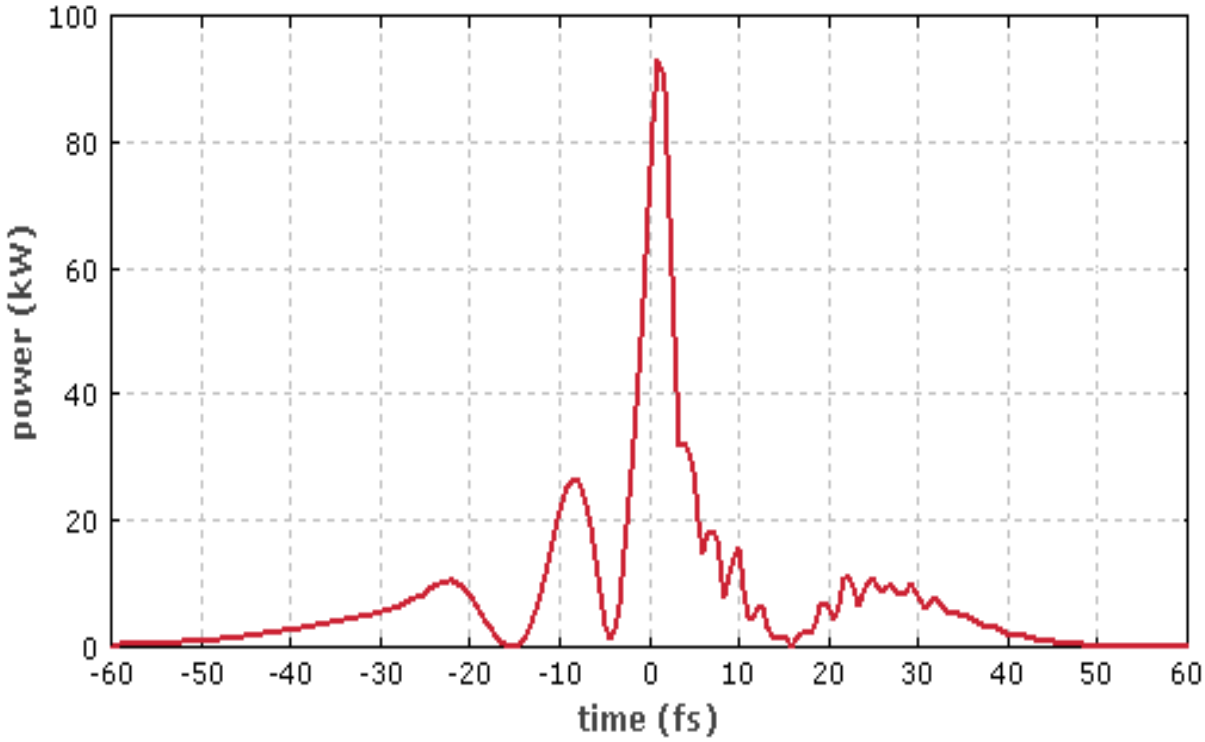

Figure 4.2: Temporal shape of a femtosecond pulse, which has been distorted during propagation in an optical fiber.

local-area networks (LANs) ,Metropolitan area networks (MANs) and in laser technology.

Secondly,rogue waves,which are an open water phenomenon, in which winds, currents, non-linear phenomena such as solitons, and other circumstances cause a wave to briefly form that is far larger than the "average" large occurring wave. The basic underlying physics that makes phenomena such as rogue waves possible is that different waves can travel at different speeds, and so they can "pile up" in certain circumstances - known as "constructive interference". (In deep ocean the speed of a gravity wave is proportional to the square root of its wavelength-the distance peak-to-peak.) However, other

situations can also give rise to rogue waves, particularly situations where nonlinear effects or instability effects can cause energy to move between waves and be concentrated in one or very few extremely large waves before returning to "normal" conditions. In such a case, it is hypothesised, an unusual, unstable wave type may form which 'sucks' energy from other waves, growing to a near-vertical monster itself, before becoming too unstable and collapsing shortly after. One simple model for this is a wave equation known as the nonlinear Schrödinger equation (NLSE), in which a normal and perfectly accountable (by the standard linear model) wave begins to 'soak' energy from the waves immediately fore and aft, reducing them to minor ripples compared to other waves. The NLS can be used in deep water conditions. In shallow water, waves are described by the Korteweg-de Vries equation or the Boussinesq equation. These equations also have non-linear contributions and show solitary-wave solutions. A small-scale rogue wave consistent with the nonlinear Schrödinger equation was produced in a laboratory water tank in 2011.[7] In particular, the study of solitons, and especially Peregrine solitons, have supported the idea that non-linear effects could arise in bodies of water.

Then, in Bose-Einstein condensate.BEC is a state of matter of a dilute gas of bosons cooled to temperatures very close to absolute zero (that is, very near 0 K or -273.16 ° C).In such situation, the movement of atom can be controlled.The most obvious property of a BEC is that a large fraction of its particles occupy the same, namely the lowest, energy state. In atomic condensates this can be confirmed by measuring the velocity distribution of the atoms in the gas. The above figure shows the result of such a measurement. In the left plot, no Bose Einstein Condensation took place. One can see that the energy distribution of the atoms is given by the Bose Einstein statistics. In the middle plot, the conditions for condensation were hardly achieved. One can still clearly see the statistically distributed atoms; but on top of this distribution a overpopulation of the ground state can also be identified, expressed by the sharp blue peak. In the picture to the right, the conditions for condensation where fulfilled very good (the temperature was lower than required), so that hardly any statistically distributed atoms can be seen. Instead one can observe a high concentration of atoms in the ground state.Using the important effects of Bose-Einstein condensate we can develop today's quantum computers that have to be cooled down slightly above the absolute zero point.[17]

In particular, when modeling a Bose-Einstein condensate with a suitably scaled interaction potential V in the large particle limit, the solution can be

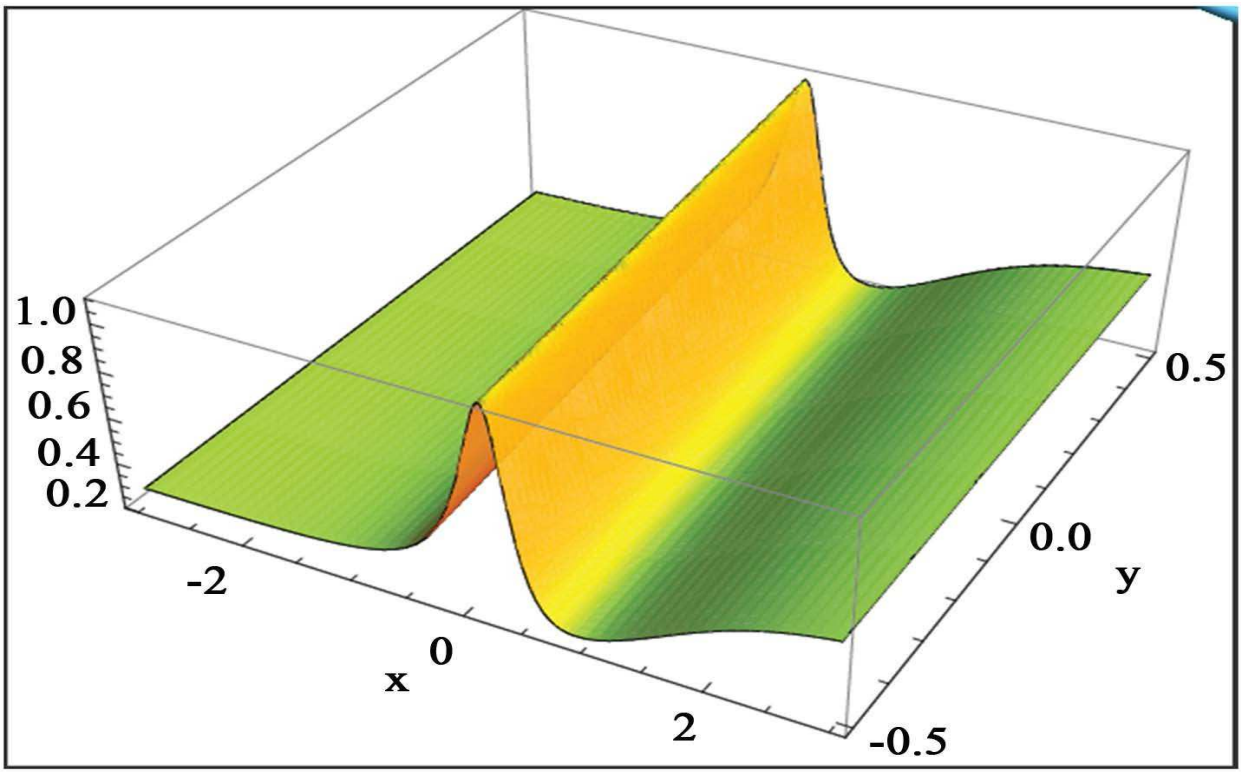

Figure 4.3: The absolute of the rogue wave profile.

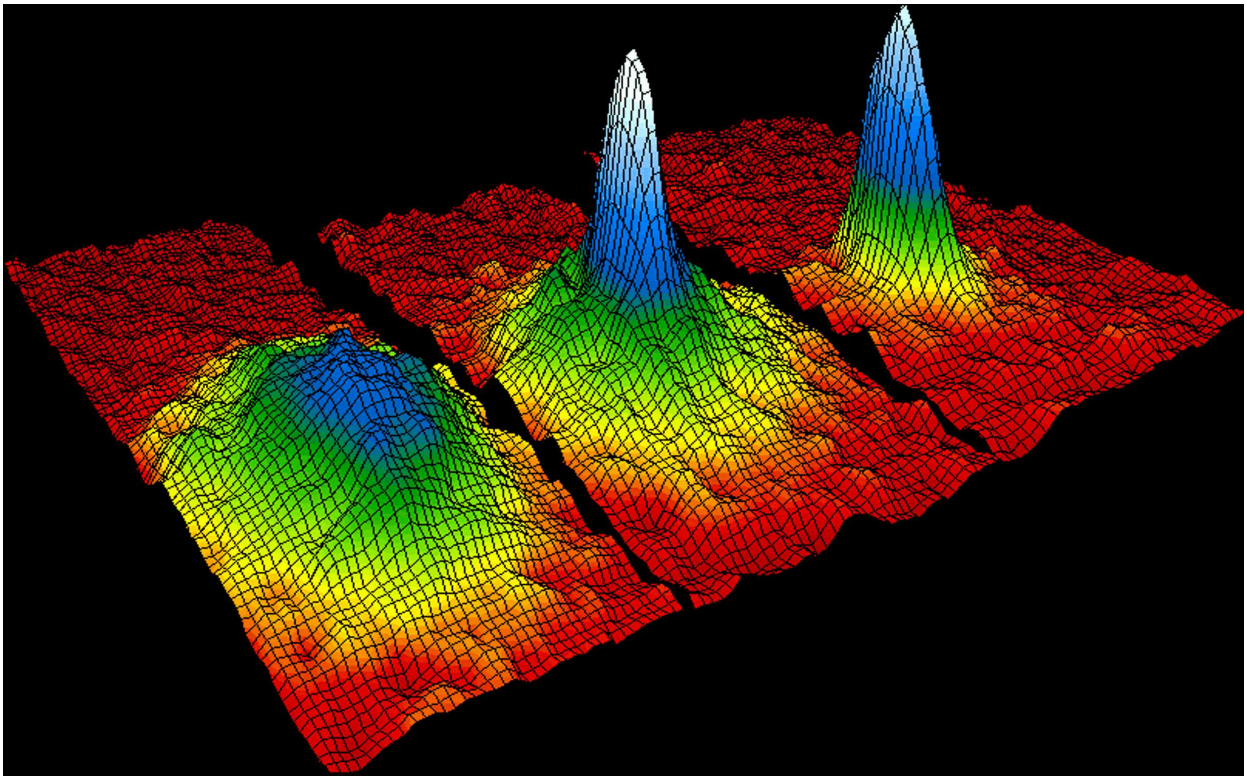

Figure 4.4: Velocity-distribution data (3 views) for a gas of rubidium atoms, confirming the discovery of a new phase of matter, the Bose-Einstein condensate.

governed by the cubic nonlinear Schrödinger equation.

In the above three specific situations,we can notice that the three figures are of same type, which can be explained using nonlinear schrödinger equation.

4.3 Finding sec hyperbolic solutions of Schrödinger with cubic non-linearity by Darboux transformation using mathematica

In the previous chapter we have discussed that for Darboux transformation method first we need Lax pair and seed solution.In chapter 3(section-3.3.4) equations 5,6,7,8,9 that we have derived to find lax pair are very important. We will rewrite these equations for convenience.
$U_{0t} - V_{0x} + [U_0, V_0] = 0$.....(a)
$U_{1t} + U_0 V_1 \psi + [U_0, V_1] + [U_1, V_0] = 0$.....(b)
$V_{2x} + [V_2, U_0] + [V_1, U_1] = 0$.....(c)
$V_{3x} + [V_3, U_0] + [V_2, U_1] = 0$.....(d)
$[V_3, U_1] = 0$.....(e)
Equation (a) gives back to the original NLSE, where $[U_0, V_0]$ is the commutator and U_0 and V_0 are the two matrices and together is the Lax pair. The other matrices U_1, V_1, V_2, V_3 in equations (b),(c),(d),(e) are the functions of the solution of the given NLSE.We can determine U_1, V_1, V_2, V_3 from U_0 and V_0 matrices.U_0 and V_0 can be found by some systematic search approach. When NLSE is of higher order we can only find U_0 and V_0, but not the other matrices.
Now, applying the systematic process we get,

$$U_0 = \begin{bmatrix} 0 & \frac{-Q[x,t]}{\sqrt{2}} \\ \frac{-Qs[x,t]}{\sqrt{2}} & 0 \end{bmatrix}$$

$$V_0 = \begin{bmatrix} \frac{1}{2} iQ[x,t]Qs[x,t] & \frac{-iQ^{(1,0)}[x,t]}{\sqrt{2}} \\ \frac{i-Qs^{(1,0)}[x,t]}{\sqrt{2}} & \frac{-1}{2} iQ[x,t]Qs[x,t] \end{bmatrix}$$

Although we have found two 2×2 matrices for U_0 and V_0, other integrability conditions do come in while deriving the remaining matrices.In the appendix [???]we will see the other matrices.Now,we will use the DT to find the analytic solution.So, we will consider the following equations.

$\Psi_x = U_0.\Psi + U_1.\Psi.\Lambda$;.....(f)
$\Psi_t = V_0.\Psi + V_1.\Psi.\Lambda + V_2.\Psi.\Lambda.\Lambda$;.....(g)

where, Ψ is the auxiliary field.And can be expressed as,

$$\Phi = \begin{bmatrix} \psi_1(x,t) & \psi_2(x,t) \\ \phi_1(x,t) & \phi_2(x,t) \end{bmatrix}$$

; Here,ψ_1,ψ_2,ϕ_1,ϕ_2 describe the auxiliary field.
In physics, and especially quantum field theory, an auxiliary field is one whose equations of motion admit a single solution.
Using this auxiliary field we find the solution of the original NLSE.
Now,applying the auxiliary field Φ on (f) and (g) we will simplify them. The equations (b),(c),(d),(e) also have to be simplified.
Here, the spectral parameter matrix Λ is written as,

$$\Lambda = \begin{bmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{bmatrix}$$

; Now, we will simplify eqx and eqt by writting in following forms, eqx = Simplify $\left[\frac{\partial \Phi}{\partial x} - \Phi \mathrm{x}\right]$
eqt = Simplify $\left[\frac{\partial \Phi}{\partial t} - \Phi \mathrm{t}\right]$;
Where, the auxiliary field is used .Then, a seed solution need to be taken for which the NLSE has to be satisfied.
Take the seed solution, $Q = 0$ and $Qs = 0$.As, we will get the elements ψ_1,ψ_2,ϕ_1 and ϕ_2 of matrix of the auxiliary field and these fields depend on some parameters c11, c12, c13, c14. These parameters are not independent and one will be changed with the change of another.
Here,U_0 contained the trivial solution of NLSE.Using this auxiliary field we find the solution $U_0[1]$ of the original NLSE, which is non-trivial and is given by,
$U_0[1] = U_0 + [U_1, \sigma]$
$U_0[1]$ is the new U_0 matrix and
here, $\sigma = \Psi_0.\Lambda.{\Psi_0}^{-1}$. So, the general solution of NLSE can be written as,

ΨN = U0 + U1.σ − σ.U1;
After simplifying this equation we get the solution,
Q(x_, t_):=$\sqrt{2}e^{it}$sech(x);
Qs(x_, t_):=$\sqrt{2}e^{-it}$sech(x);
which is a sec hyperbolic solution.
For the parameters c11 = −1; c12 = 1; c13 = −1; c14 = −1; λ1 = 1; $and\lambda$2 = −1; we get this solution.For these parameters and different time,t=0, t=5, t=10, t=15 we get the following figures,

4.3.1 Discussions1

From the figures above we have shown the evolution of soliton wave with position and we can see that for different times the shape and size of the wave or wave function is not changing. And this is the property of an usual fundamental soliton wave that can maintain its profile unchanged during propagation.As,the solution is a hyperbolic function, so the properties of the wave is exactly same as sec hyperbolic function.That is, the profile(wave function) will be symmetric everywhere and everytime(i.e; for all position and time).
Here, we can clearly notice the effect of very strong nonlinear term is this wave function. The wave function is not changing over time,which is the main property of a soliton.The reason why it is happening to the wave is that the dispersive term of the wave is being cancel out by the nonlinear term(as dispersive termis trying to spread the wave and vanish it, while nonlinear term is trying to suck up all the energies frond surrounding and increase the peak height).That's why both term(dispersive and nonlinear) being canceled out by each other can let the wave go a long way maintaining its profile.
In these figure we have shown the change of wave function ψ(x,t) with x(-10 to 10) for different time t=0,5,10,15. Here the wave profile is changing between 0 to 2.

We can get other solutions by changing the parameters.
Now for parameters, c11=-0.5; c12=0.5; c13=-0.5; c14=-0.5; λ1=1; and λ2=-1; we get the following solution,
Q(x_, t_):=2.sech(x)(cos(t) + (0. + 1.i) sin(t))
which is also a sec hyperbolic solution.So, we will get same kind of time evolution as before.
In this case for different times we get the following figures,

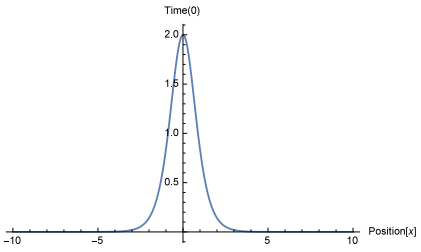

Figure 4.5: For time t=0.

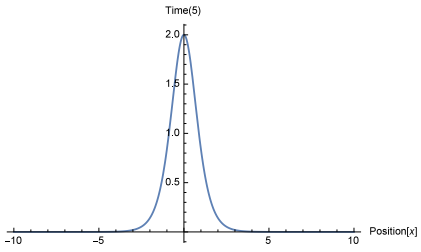

Figure 4.6: For time t=5.

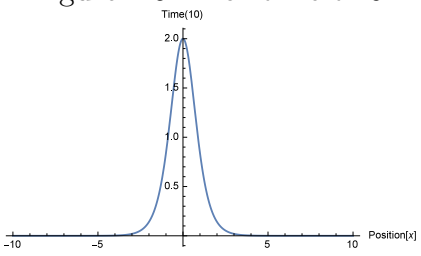

Figure 4.7: For time t=10.

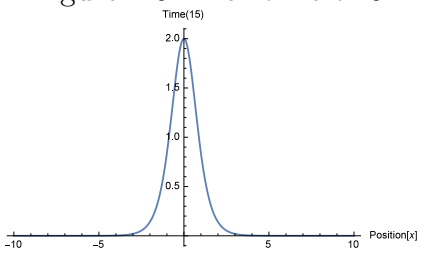

Figure 4.8: For time t=15.

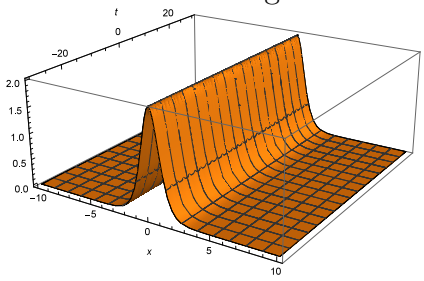

Figure 4.9: 3D plot for any time(in general).

Figure 4.10: Graphs for the sec hyperbolic solution of nonlinear Schrödinger equation for different times,For parameters c11=-1;c12=1; c13=-1; c14=-1;λ1=1; and λ2=-1;

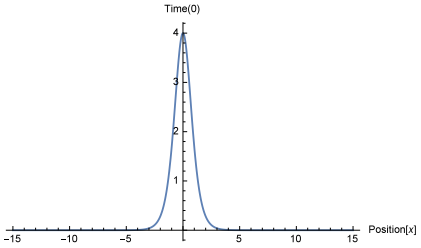

Figure 4.11: For time t=0.

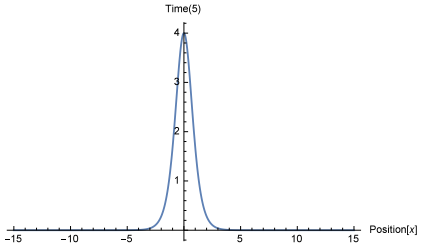

Figure 4.12: For time t=5.

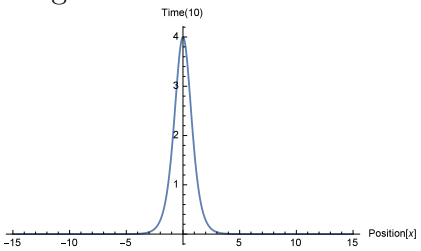

Figure 4.13: For time t=10.

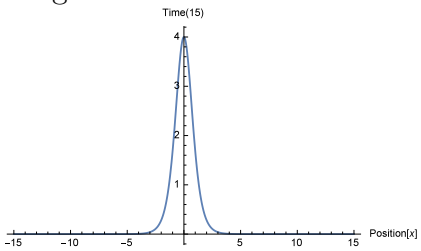

Figure 4.14: For time t=15.

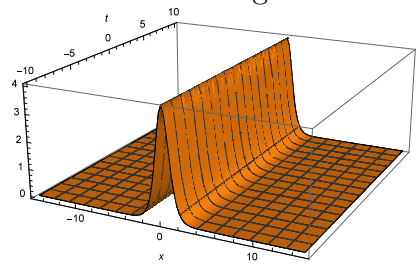

Figure 4.15: 3D plot for any time(in general).

Figure 4.16: Graphs for sec hyperbolic solution of nonlinear Schrödinger equation for different times,For parameters c11=-0.5;c12=0.5; c13=-0.5; c14=-0.5;λ1=1; and λ2=-1;

4.3.2 Discussions2

Here, after changing the several parameters we also get the sec hyperbolic solution.This is almost same as we discussed in Discussions1. Here the only difference is that the wave profile is changing from 0 to 5 with position x.
From the above two sec hyperbolic solutions we get to know that the time evolution of the wave function is similar for all time.Though the time is increasing, the height and shape of the wave is remaining same and the sec hyperbolic solutions has changed with the change of the parameters.An example is given which shows that solution is dependent on parameters and the parameters are also mutually dependent.Here, 3D figures are also shown to clear the visual concept.

4.4 Example:

Lets start with a time dependent nonlinear Schrödinger equation:
$-i\frac{\partial}{\partial t}\psi = \frac{1}{2}\frac{\partial^2\psi}{\partial x^2} + \frac{1}{2}k|\psi|^2\psi$
where k is a parameter.
Taking a trial $\psi = 0$ we get the following nontrivial sec hyperbolic solution,
$\psi = Asech(bx)e^{iat}$ we get,[where, A and b are parameters]
$-i\frac{\partial}{\partial t}Asech(bx)e^{iat} = -\frac{1}{2}\frac{\partial^2}{\partial x^2}(Asech(bx)e^{iat}) + \frac{1}{2}k(Asech(bx)e^{iat}.Asech(bx)e^{-iat}).Asech(bx)e^{iat}$
$\Rightarrow -i(\frac{\partial}{\partial t}e^{iat}).Asech(bx) = -\frac{1}{2}Ae^{iat}\frac{\partial^2}{\partial x^2}(sech(bx)) + \frac{1}{2}k(A^3sech^3(bx).e^{iat})$
$\Rightarrow -i(ia)e^{iat}Asech(bx) = -\frac{1}{2}Ae^{iat}\frac{\partial^2}{\partial x^2}sech(bx) + \frac{1}{2}kA^3e^{iat}sech^3(bx)$.....(3.7)
Now,
$\frac{\partial}{\partial x}(sech(bx)) = -sech(bx)tanh(bx).b$
$\frac{\partial^2}{\partial x^2}(sech(bx)) = -b\frac{\partial}{\partial x}(sech(bx)tanh(bx))$
$\Rightarrow \frac{\partial^2}{\partial x^2}(sech(bx)) = -b[sech(bx)\frac{\partial}{\partial x}tanh(bx) + tanh(bx)\frac{\partial}{\partial x}sech(bx)]$
$\Rightarrow \frac{\partial^2}{\partial x^2}(sech(bx)) = -b[sech(bx)(sech^2(bx)).b + tanh(bx)tanh(bx).sec(bx).b]$
$\Rightarrow -b^2[sech^3(bx) - sech(bx).tanh^2(bx)]$
From equation (3.7) we get,
$-i(ia)e^{iat}Asech(bx) = -\frac{1}{2}Ae^{iat}[-b^2[(sech^3(bx) - sech(bx)tanh^2(bx))]] + \frac{1}{2}kA^3e^{iat}sech^3(bx)$
$\Rightarrow asech(bx) = -\frac{b^2}{2}sech(bx)[sech^2(bx) - tanh^2(bx)] + \frac{1}{2}kA^2sech^2(bx)sech(bx)$
$\Rightarrow asech(bx) = -\frac{b^2}{2}[\frac{1}{cosh^3(bx)} - \frac{sinh^2(bx)}{cosh^3(bx)}] + \frac{1}{2}kA^2\frac{1}{cosh^3(bx)}$
$\Rightarrow asech(bx) = -\frac{b^2}{2}[\frac{1-sinh^2(bx)}{cosh^3(bx)}] + \frac{1}{2}\frac{kA^2}{cosh^3(bx)}$
$\Rightarrow asech(bx) = -\frac{b^2}{2}sech^3(bx)[1 - sinh^2(bx)] + \frac{kA^2}{cosh^3(bx)}$

$\Rightarrow asech(bx) = -\frac{b^2}{2}sech^3(bx)[1 - cosh^2(bx) + 1] + kA^2$
$\Rightarrow asech(bx) = -\frac{b^2}{2}sech^3(bx)[-cosh^2(bx) + 2 + kA^2]$
$\Rightarrow a = -\frac{1}{2}sech^2(bx)[-cosh^2(bx)b^2 + 2b^2 + kA^2]$
Here, a,b,k are parameters which are dependent on each other, where for $a = -\frac{b^2}{2}$ and $A^2 = -\frac{2b^2}{k}$,$\psi = Asech(bx)e^{iat}$ is a nontrivial solution of the Nonlinear Schrödinger equation with cubic non-linearity.

4.5 Finding peregrine sotutions of Schrödinger with cubic non-linearity by Darboux transformation using mathematica

4.5.1 Peregrine Soliton

The Peregrine soliton (or Peregrine breather) is an analytic solution of the nonlinear Schrödinger equation [16]. This solution has been proposed in 1983 by Howell Peregrine, researcher at the mathematics department of the University of Bristol.

Main properties

Contrary to the usual fundamental soliton that can maintain its profile unchanged during propagation, the Peregrine soliton presents a double spatio-temporal localization. Therefore, starting from a weak oscillation on a continuous background, the Peregrine soliton develops undergoing a progressive increase of its amplitude and a narrowing of its temporal duration. At the point of maximum compression, the amplitude is three times the level of the continuous background (and if one considers the intensity as it is relevant in optics, there is a factor 9 between the peak intensity and the surrounding background). After this point of maximal compression, the wave's amplitude decreases and its width increases and it finally vanishes.

These features of the Peregrine soliton are fully consistent with the quantitative criteria usually used in order to qualify a wave as a rogue wave.

Therefore, the Peregrine soliton is an attractive hypothesis to explain the formation of those waves which have a high amplitude and may appear from nowhere and disappear without a trace [21].

4.5.2 Solution

In case of peregrine soliton solution we will take different seed solution. The seed solution is,
$Q = e^{i}t$ and $Qs = e^{-}it$
Here, after applying the Darboux transformation we will get the following solutions for parameters λ1=1, λ2=-1, f11 = 1; f110 = 2; g11 = 0; c4(t) = -t; a1(t) = 2; a2(t) = 1.
$Q(x,t) = -\frac{4e^{2t^2+it(x+2)+x}}{e^{4t^2}+e^{2x}}$
For this solution we get the following figures,

4.5.3 Discussions3

Here. in the figure above we can see that wave profile is propagating and changing with time, which is not the property of usual soliton. With time, the intensity of the wave is increasing if we notice figures(4.17 to 4.22).It will keep increasing the pick height as the wave travels through the medium until defocusing effects or medium damage interrupt this process, which is not shown in the figures. This is the property of self-focusing wave.As peregrine solitons are created suddenly and remain some time and then vanish,the same occurs to this wave.
Here, a 3D figure is also shown to clear the visual concept.

For the same seed solution and changing the parameters to we get the following solution.

Figure: Graphs for the defocusing peregrine soliton solution of nonlinear Schrödinger equation for different times.

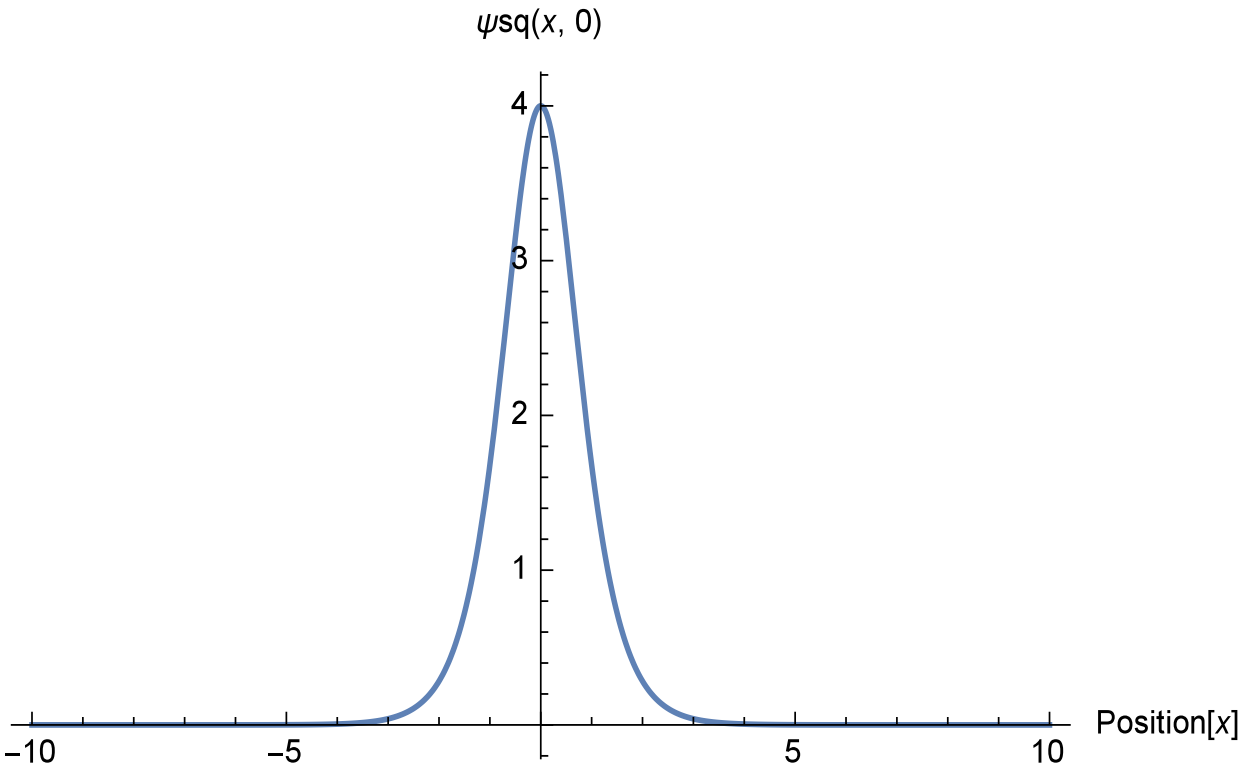

Figure 4.17: For time t=0 and x(-10 to 10)(s1)

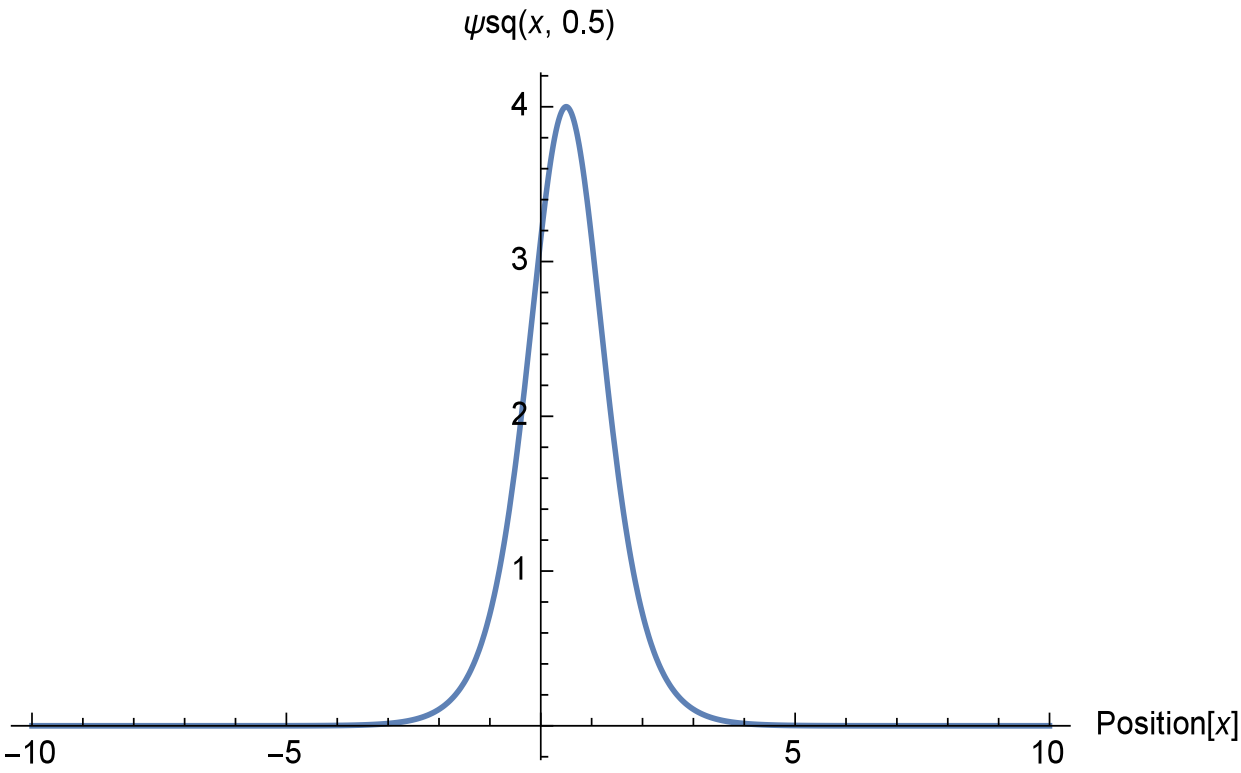

Figure 4.18: For time t=0.5

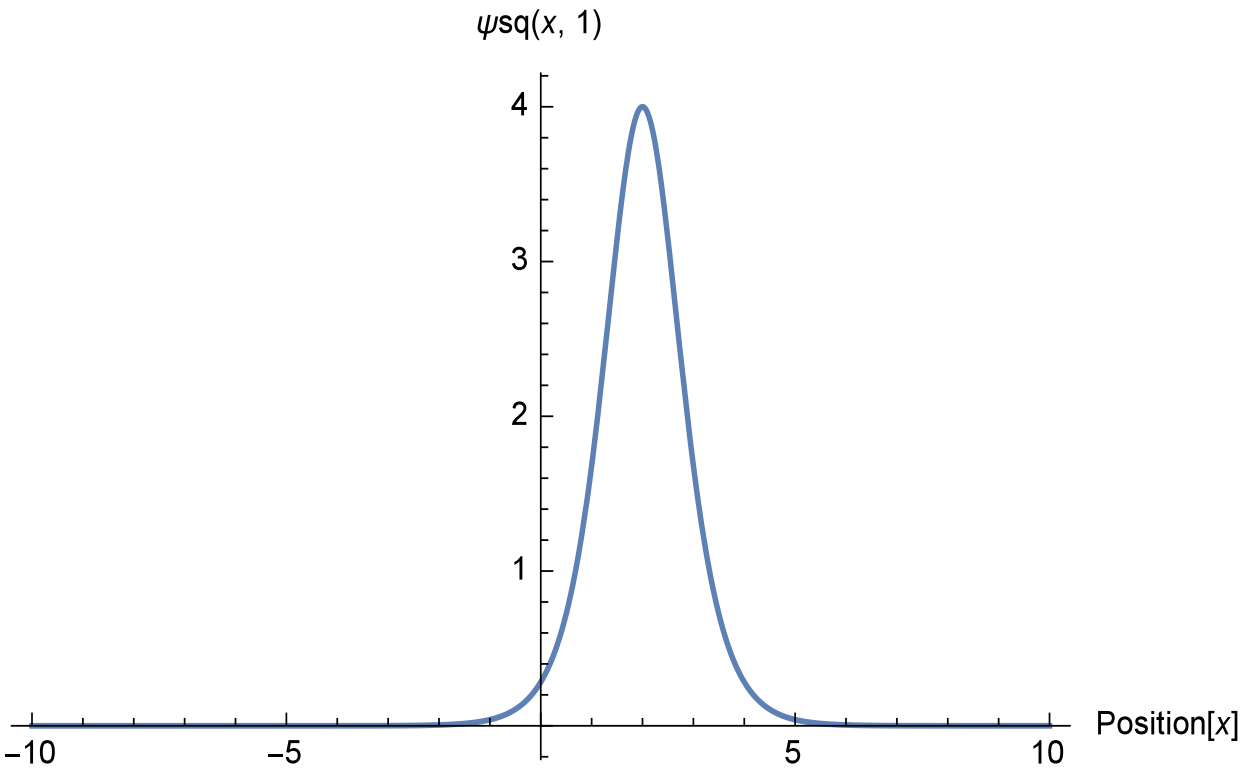

Figure 4.19: For time t=1.

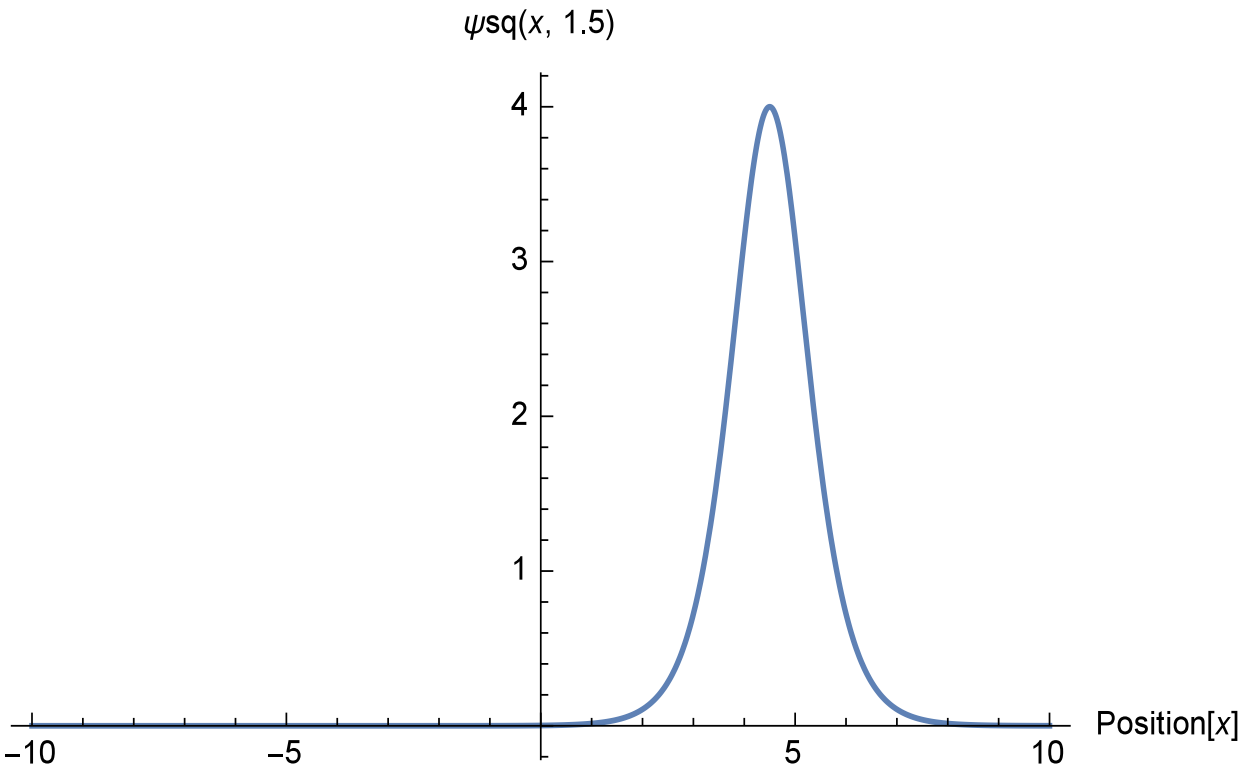

Figure 4.20: For time t=1.5

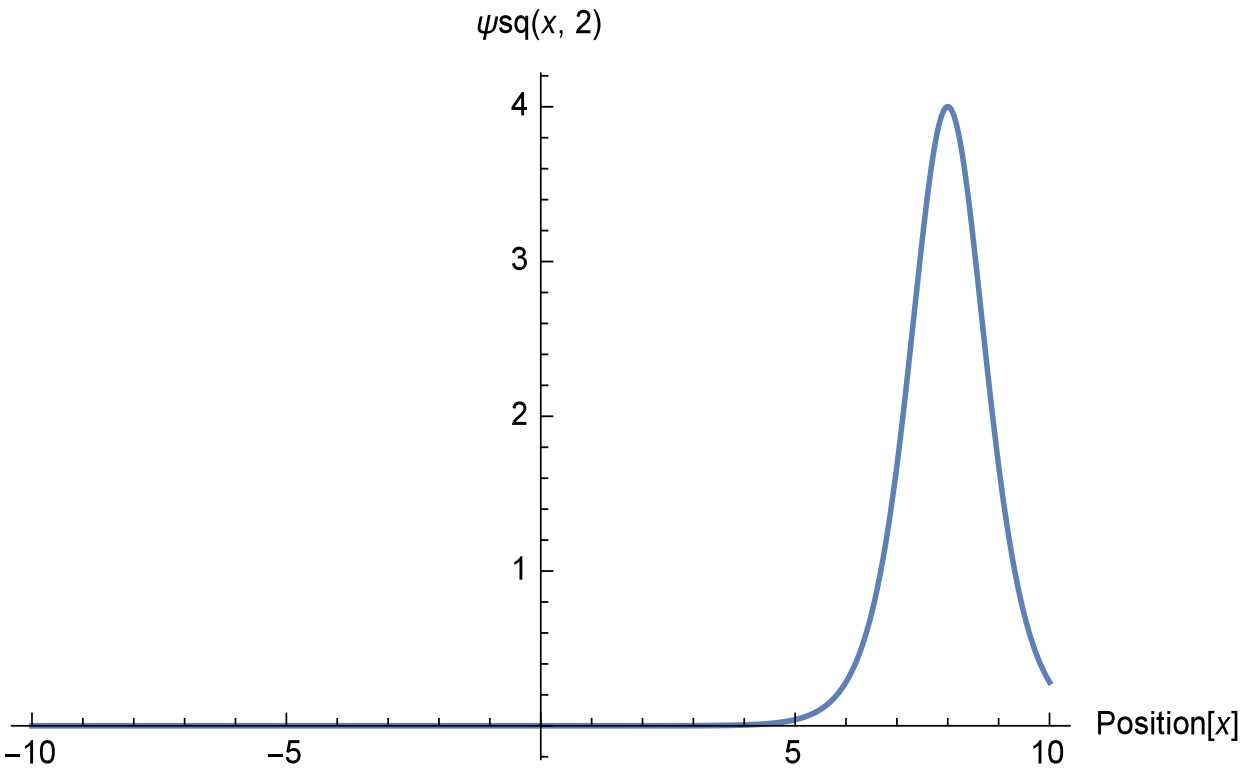

Figure 4.21: For time t=2.

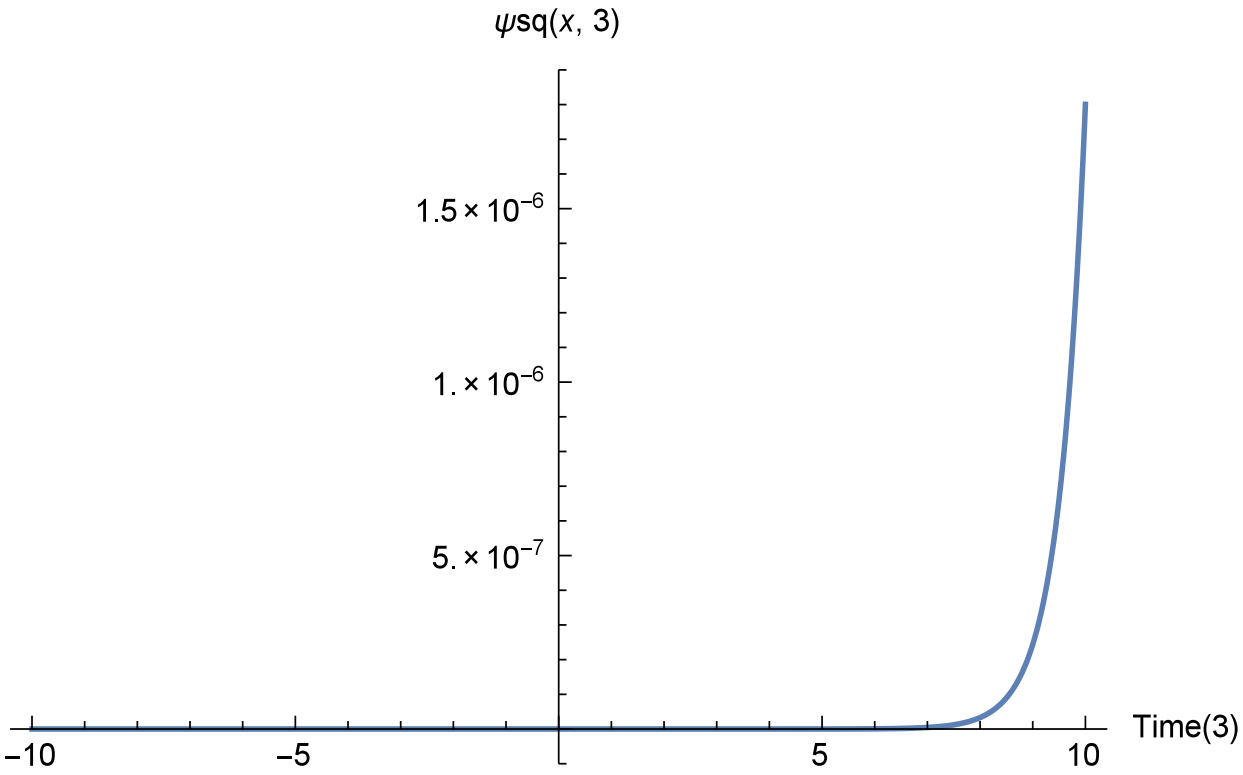

Figure 4.22: For time t=3.

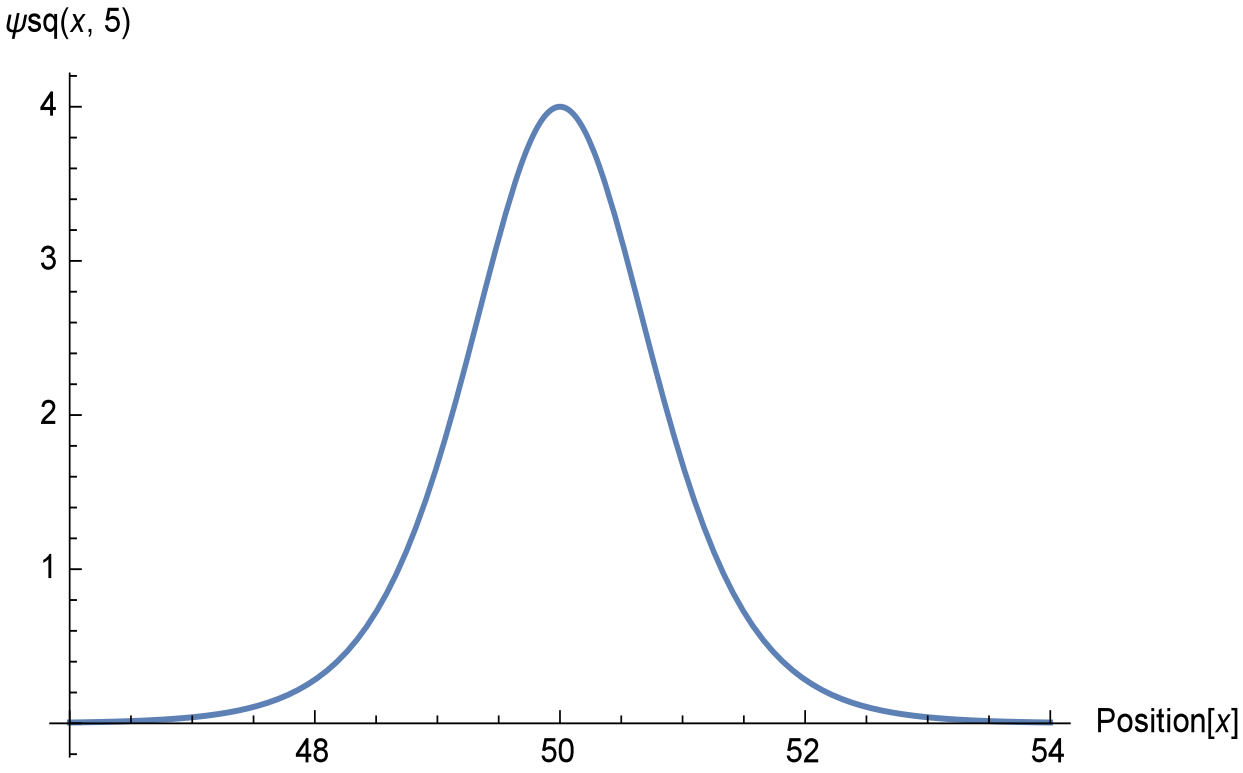

Figure 4.23: For time t=5 and x=(46 to 54)(s2).

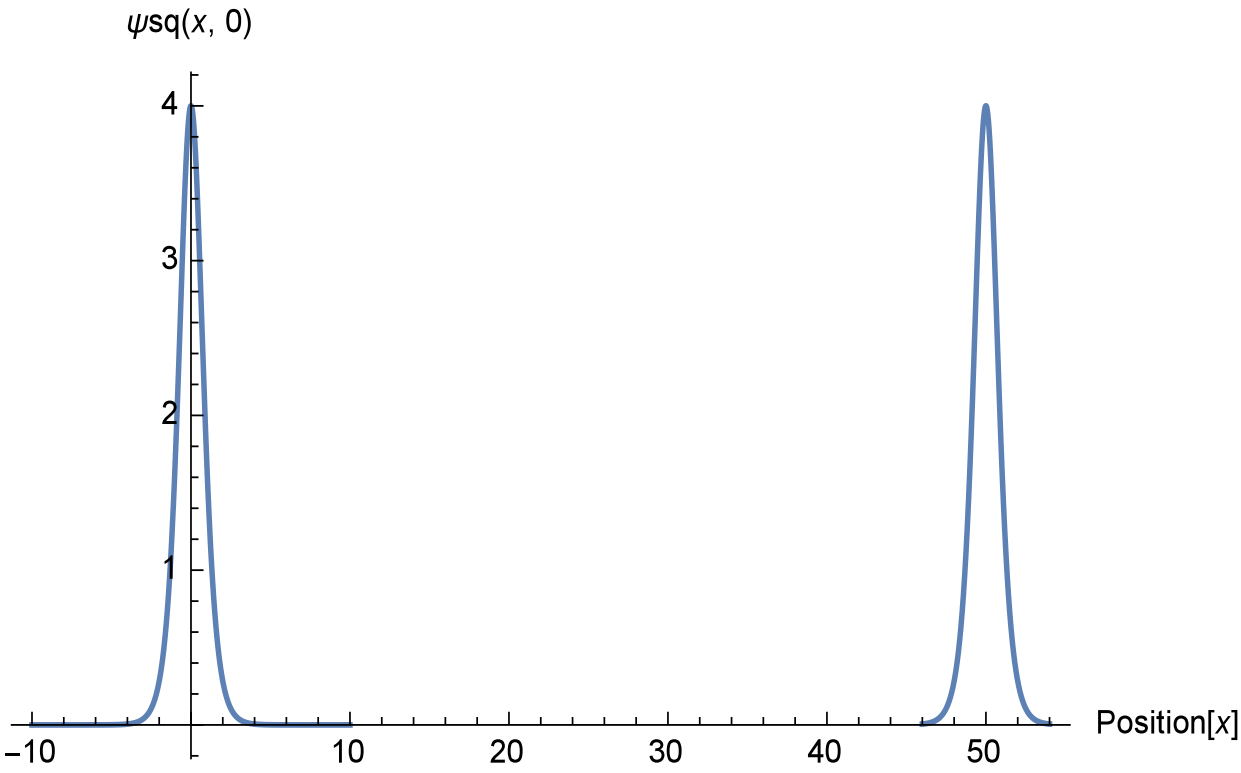

Figure 4.24: (s1) and (s2)

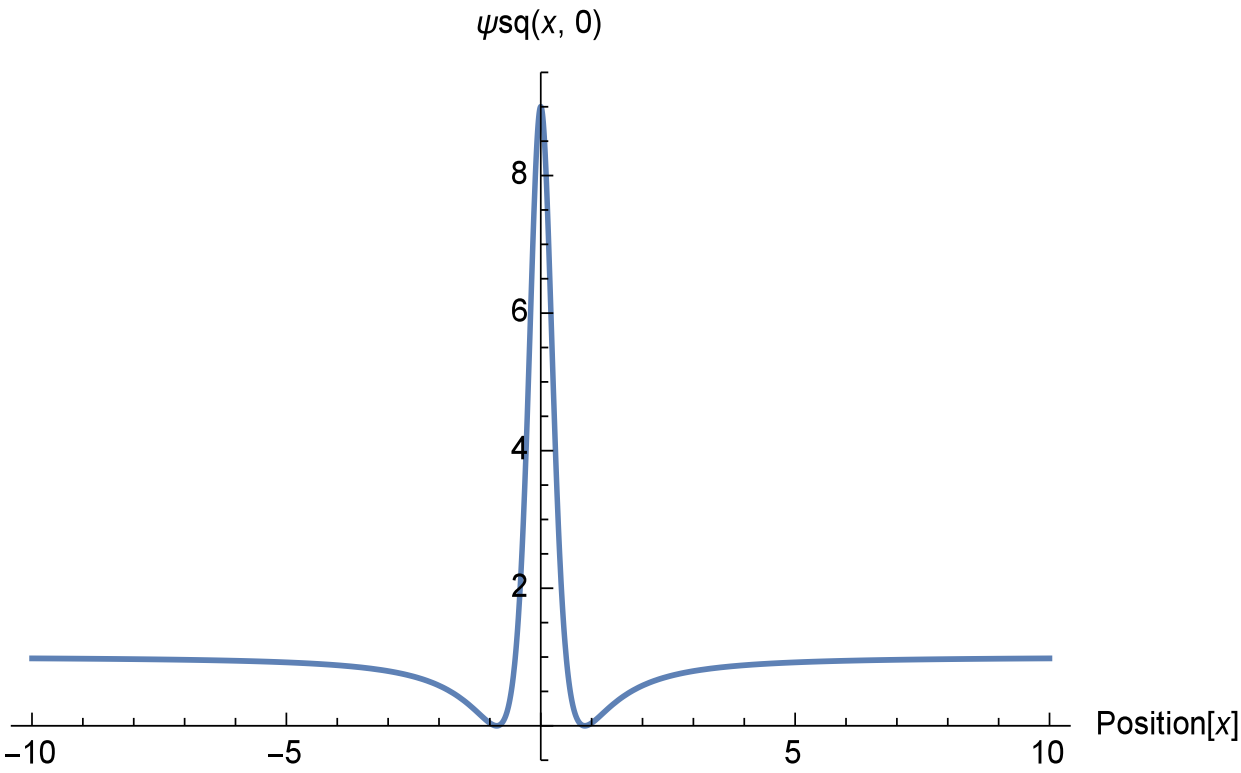

Figure 4.25: For time t=0.

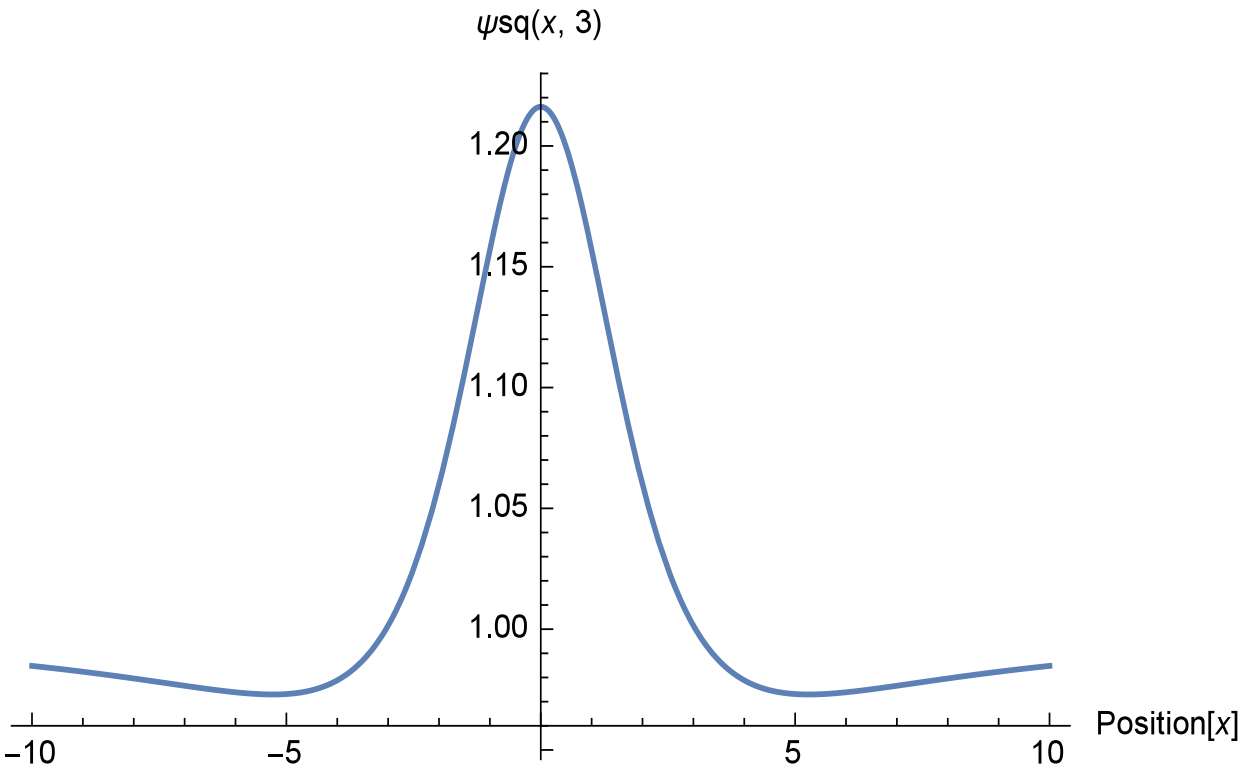

Figure 4.26: For time t=3

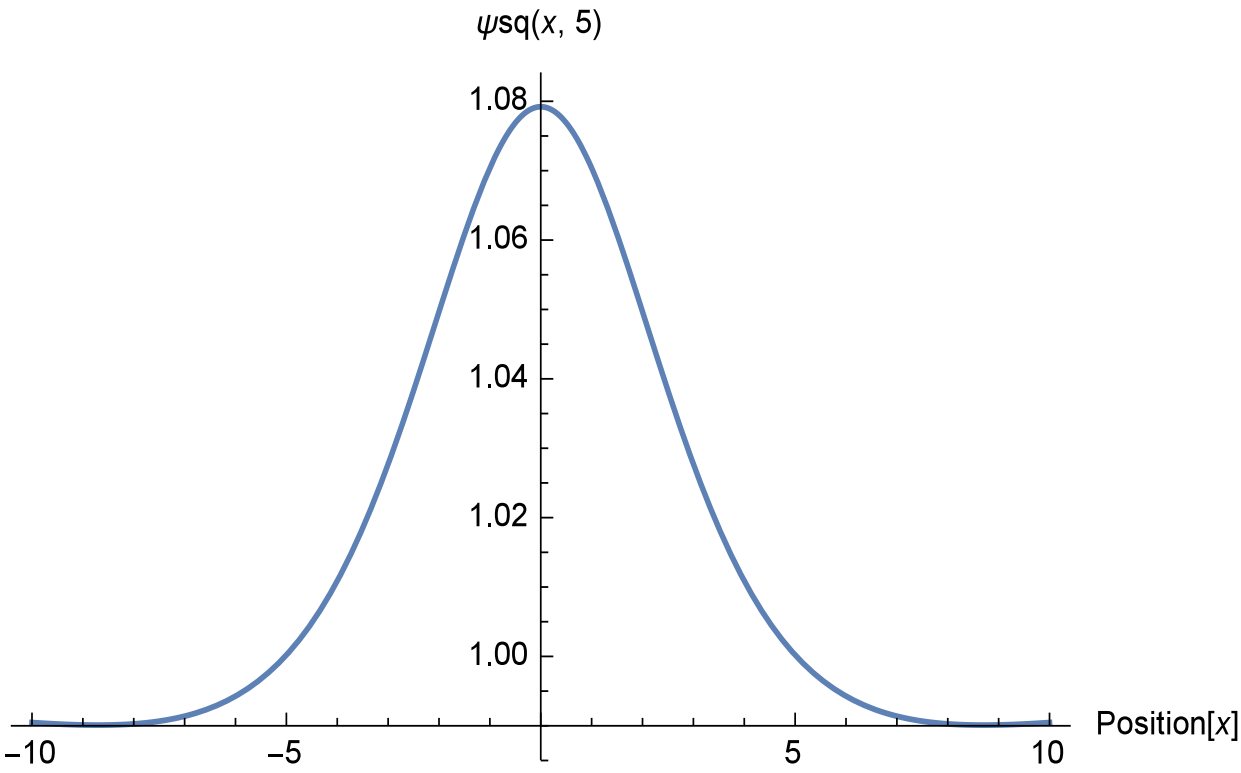

Figure 4.27: For time t=5

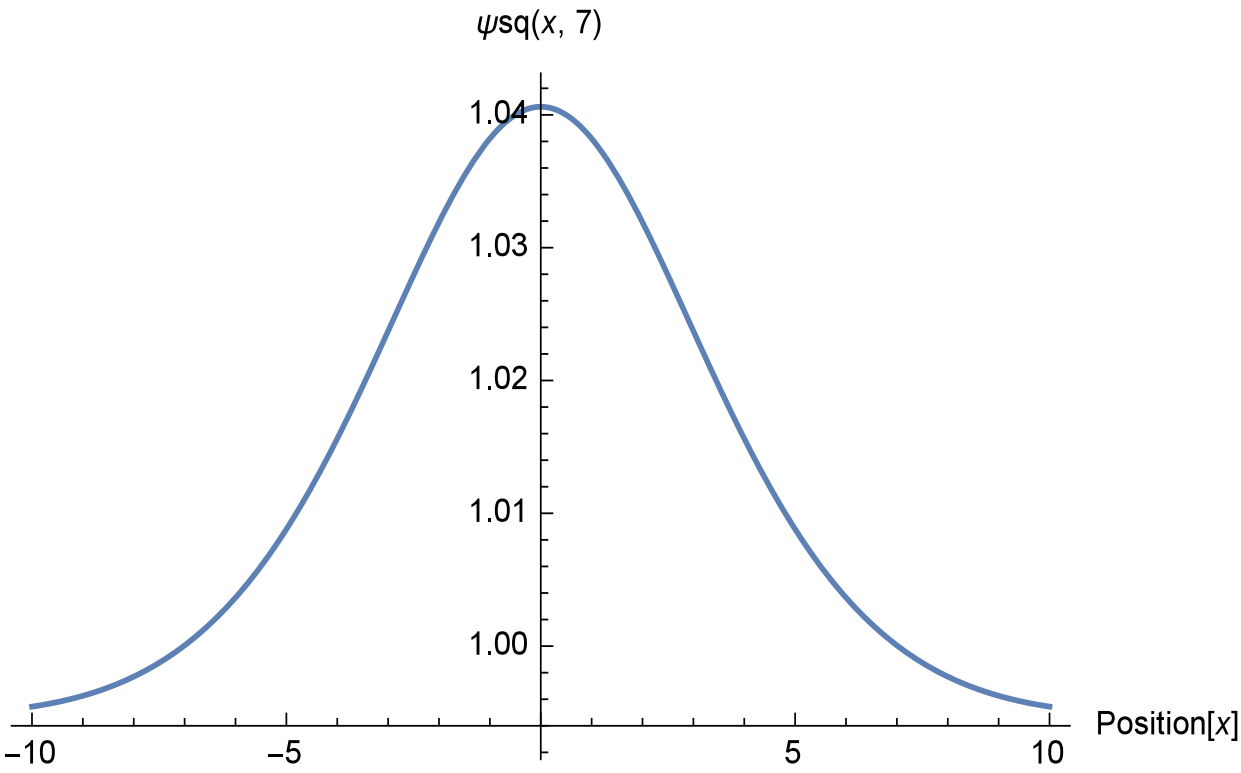

Figure 4.28: For time t=7.

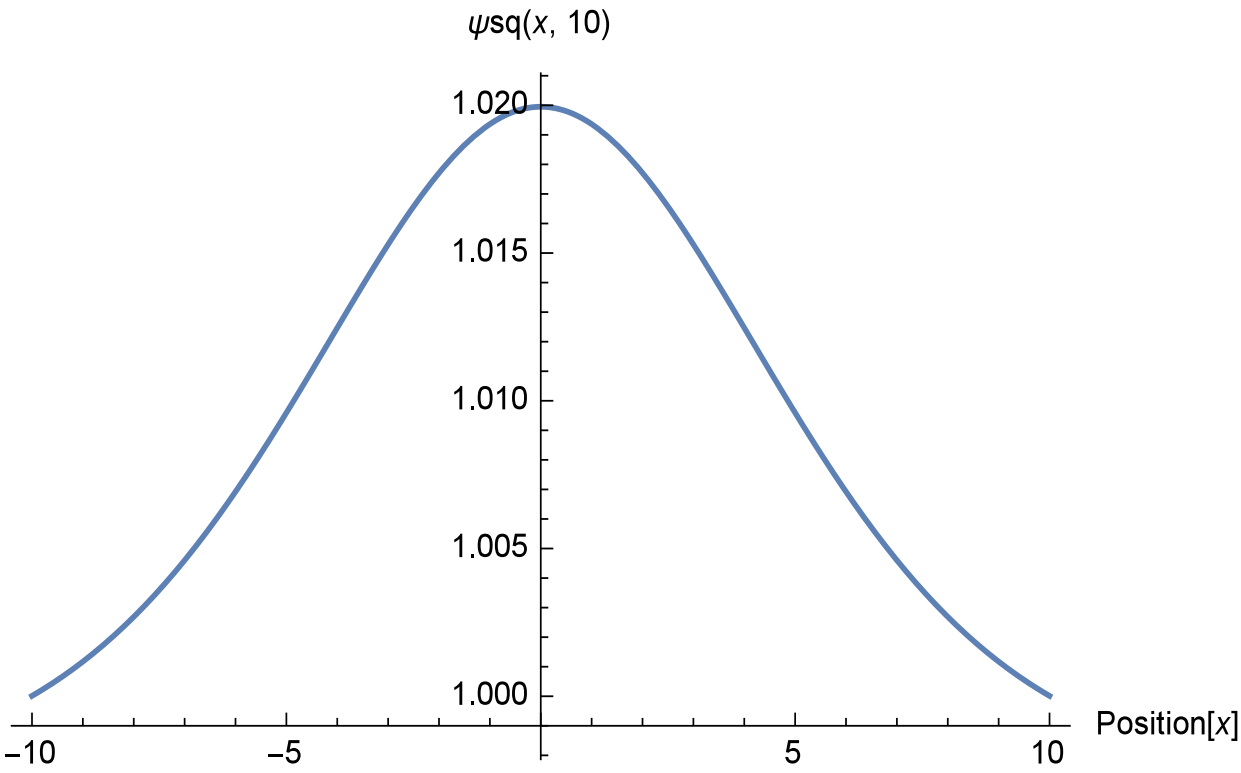

Figure 4.29: For time t=10.

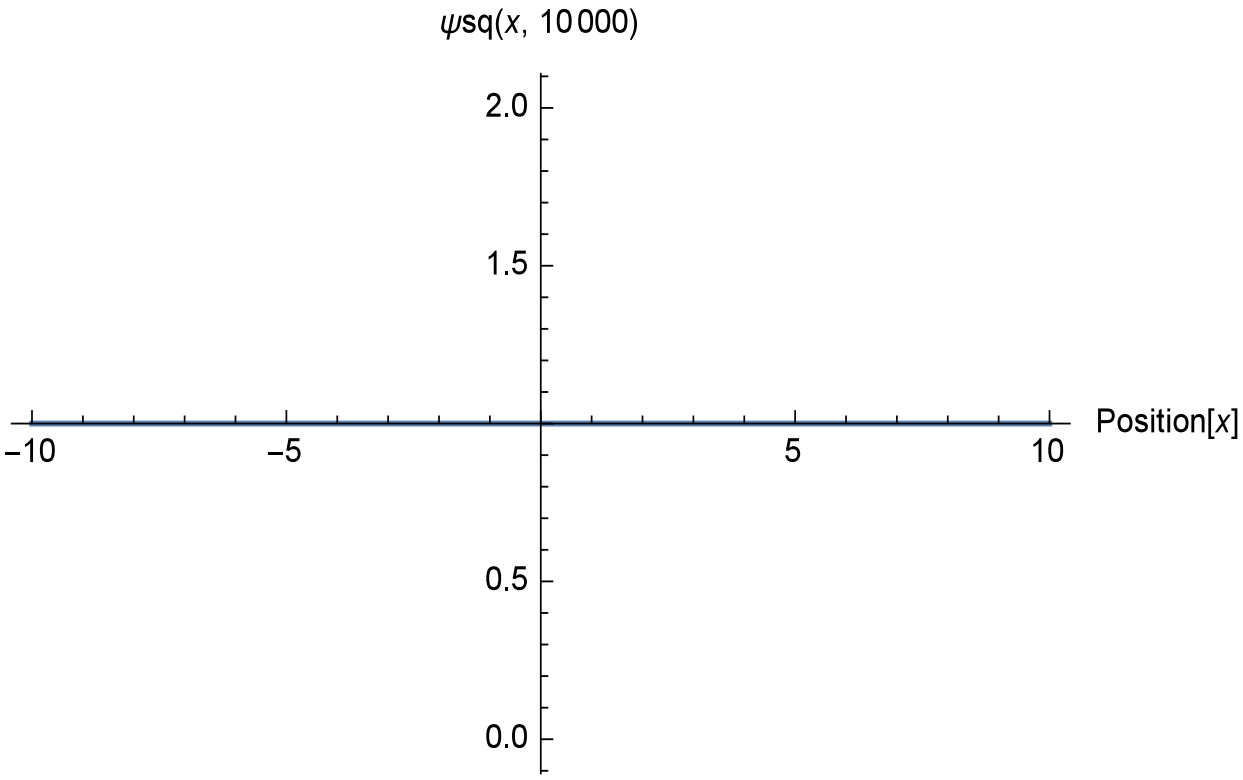

Figure 4.30: For time t=1000

4.5.4 Discussions4

Here, in the figures(4.27 to 4.32) we can see that for time t=0 we get a wave profile with intense peak.After increasing the time the wave profiles' amplitude is decreasing and width is increasing and finally in figure-4.32 the wave just has vanished.This processing called defocusing. So. these figures represents defocusing peregrine soliton.
Here, a 3D figure is also shown to clear the visual concept.

Chapter 5

Conclusion

5.1 General Conclusion

We have derived the Lax pairs for the general NLSEs with cubic and and quintic non-linearities using a systematic search approach. The existence of the Lax pairs have been verified by a similarity transformation. From the Lax pair for the general HNLSE with cubic non-linearity we are able to get the Lax pairs for some of the most well-known NLSEs, namely, Hirota, Derivative nonlinear Shrödinger, Sasa-Satsuma, and Gross-Pitaevskii equations, in addition to new ones. These equations are considered to be special cases of the general HNLSE with cubic non-linearity.

We are also able to get the Lax pair for the general HNLSE with cubic non-linearity from the Lax pair of the general HNLSE with cubic and quintic non-linearities. The former equation is a special case of the later equation when the quintic terms are zero. New exact solutions of the HNLSE can be found and the nonlinear initial-value problem can be solved using the Lax pairs found here through the Darboux transform. From the general solutions of HNLSEs, one will be able to get the solutions of the other NLSEs by simply changing the parameter set.

5.2 Specific Conclusion

Though there are many other methods for solving NLSE analytically and numerically we used analytical approach, because analytical approaches are more accurate than numerical approaches.Numerically sometimes we find some result which are partially possible or true.That's why used analytical approaches.And among other analytical approaches Darboux Transformation

is more easier and reliable.

We have got sec hyperbolic and peregrine solutions for different seed solutions and Lax pairs. After changing parameters we found that the solution has changed a little bit.

5.3 Future Possibilities

Though we are working on this, still we dont know how to control natural phenomena like natural disasters.We hope in future some new ideas will be revealed so that we can control our nature. There have been found some non-linear facts in astronomy like, in infrared astronomical satellite (IRAS) galaxy samples,Black hole x-ray nova,Galactic black hole(GBH) light curves etc.[9].So, we hope one day we will be able to solve these nonlinear problems and can acquire deeper knowledge of science.The universe will be more familier to us then it is now.

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Appendix A

Finding sec hyperbolic solution for Nonlinear Schrödinger equation using Darboux Transformation Method

$$U0 = \begin{bmatrix} 0 & \frac{-Q[x,t]}{\sqrt{2}} \\ \frac{-Qs[x,t]}{\sqrt{2}} & 0 \end{bmatrix}$$

$$V0 = \begin{bmatrix} \frac{1}{2}iQ[x,t]Qs[x,t] & \frac{-iQ^{(1,0)}[x,t]}{\sqrt{2}} \\ \frac{i-Qs^{(1,0)}[x,t]}{\sqrt{2}} & \frac{-1}{2}iQ[x,t]Qs[x,t] \end{bmatrix}$$

$$U1 = \begin{bmatrix} \frac{1}{2} & 0 \\ 0 & \frac{-1}{2} \end{bmatrix}$$

$$V1 = \begin{bmatrix} 0 & \frac{iQ[x,t]}{\sqrt{2}} \\ -\frac{iQ[x,t]}{\sqrt{2}} & 0 \end{bmatrix}$$

$$V2 = \begin{bmatrix} \frac{i}{2} & 0 \\ 0 & -\frac{i}{2} \end{bmatrix}$$

```
    eq1 = D[U0, t] - D[V0, x] + U0.V0 - V0.U0;
eq2 = D[U1, t] - D[V1, x] + U0.V1 - V1.U0 + U1.V0 - V0.U1;

    Simplify[U0] // MatrixForm
Simplify[V0] // MatrixForm

    Simplify[eq1] // MatrixForm
Simplify[eq2] // MatrixForm

    Simplify[U1] // MatrixForm
```

```
Simplify[V1] // MatrixForm
Simplify[V2] // MatrixForm
```

```
    Simplify[D[U0, t] - D[V0, x] + U0.V0 - V0.U0] // MatrixForm
Simplify[D[U1, t] - D[V1, x] + U0.V1 - V1.U0 + U1.V0 - V0.U1] // Matrix-
Form
Simplify[D[V2, x] + V2.U0 - U0.V2 + V1.U1 - U1.V1] // MatrixForm
Simplify[U1.V2 - V2.U1] // MatrixForm
```

$$\Lambda = \begin{bmatrix} \lambda 1 & 0 \\ 0 & \lambda 2 \end{bmatrix}$$

$$\Phi = \begin{bmatrix} \psi 1(x,t) & \psi 2(x,t) \\ \phi 1(x,t) & \phi 2(x,t) \end{bmatrix}$$

```
    Φx=U0.Φ + U1.Φ.Λ ;
Φt=V0.Φ + V1.Φ.Λ + V2.Φ.Λ.Λ;
```

```
    Simplify[eqx] // MatrixForm
Simplify[eqt] // MatrixForm
```

```
    Q[x,t] := 0
Qs[x,t] := 0
Simplify[eqx] // MatrixForm
Simplify[eqt] // MatrixForm
```

```
    eq1x = eqx[[1, 1]];
eq2x = eqx[[1, 2]];

    eq3x = eqx[[2, 1]];
eq4x = eqx[[2, 2]];

    eq1t = eqt[[1, 1]];
eq2t = eqt[[1, 2]];
```

eq3t = eqt[[2, 1]];
eq4t = eqt[[2, 2]];

Simplify[eq1x]
Simplify[eq2x]

Simplify[eq3x]
Simplify[eq4x]

Simplify[eq1t]
Simplify[eq2t]

Simplify[eq3t]
Simplify[eq4t]

$ReplaceAll[DSolve[eq1x == 0, \Psi 1[x,t], x], K[1] \to x]$
$ReplaceAll[DSolve[eq2x == 0, \Psi 2[x,t], x], K[1] \to x]$
$ReplaceAll[DSolve[eq3x == 0, \Phi 1[x,t], x], K[1] \to x]$
$ReplaceAll[DSolve[eq4x == 0, \Phi 2[x,t], x], K[1] \to x]$

$\Psi 1[x_, t_] := c11[t]e^{-\frac{1}{2}(\lambda 1 x)}$
$\psi 2$(x_, t_):=c12$(t)e^{-\frac{1}{2}(\lambda 2 x)}$
$\phi 1$(x_, t_):=c13$(t)e^{\frac{\lambda 1 x}{2}}$
$\phi 2$(x_, t_):=c14$(t)e^{\frac{\lambda 2 x}{2}}$

Simplify[eq1x]
Simplify[eq2x]

Simplify[eq3x]
Simplify[eq4x]

Simplify[eq1t]
Simplify[eq2t]

Simplify[eq3t]
Simplify[eq4t]

ReplaceAll[DSolve[eq1t == 0, c11[t], t], K[1] → t]
ReplaceAll[DSolve[eq2t == 0, c12[t], t], K[1] → t]
ReplaceAll[DSolve[eq3t == 0, c13[t], t], K[1] → t]
ReplaceAll[DSolve[eq4t == 0, c14[t], t], K[1] → t]

$\psi 1(\mathrm{x_}, \mathrm{t_}) := \mathrm{c11} e^{\frac{1}{2} i \lambda 1^2 t} e^{-\frac{1}{2}(\lambda 1 x)}$
$\psi 2(\mathrm{x_}, \mathrm{t_}) := \mathrm{c12} e^{\frac{1}{2} i \lambda 2^2 t} e^{-\frac{1}{2}(\lambda 2 x)}$
$\phi 1(\mathrm{x_}, \mathrm{t_}) := \mathrm{c13} e^{-\frac{1}{2} i \lambda 1^2 t} e^{\frac{\lambda 1 x}{2}}$

$\phi 2(\mathrm{x_}, \mathrm{t_}) := \mathrm{c14} e^{-\frac{1}{2} i \lambda 2^2 t} e^{\frac{\lambda 2 x}{2}}$

Simplify[eq1x]
Simplify[eq2x]

Simplify[eq3x]
Simplify[eq4x]

Simplify[eq1t]
Simplify[eq2t]

Simplify[eq3t]
Simplify[eq4t]

$Simplify[\Psi 1[x, t]]$
$Simplify[\Psi 2[x, t]]$

$Simplify[\Phi 1[x, t]]$
$Simplify[\Phi 2[x, t]]$

Simplify[eq1x]
Simplify[eq2x]

Simplify[eq3x]
Simplify[eq4x]

Simplify[eq1t]
Simplify[eq2t]

Simplify[eq3t]
Simplify[eq4t]

$Q[x,t] := 0$
$Qs[x,t] := 0$

$\Psi 1[x,t] := c11E^(1/2I\Lambda 1(Ix + t\Lambda 1))$
$\Psi 2[x,t] := c12E^(1/2I\Lambda 2(Ix + t\Lambda 2))$

$\Phi 1[x,t] := c13E^(1/2\Lambda 1(x - It\Lambda 1))$

$\Phi 2[x,t] := c14E^(1/2\Lambda 2(x - It\Lambda 2))$

$$\Phi = \begin{bmatrix} \psi 1(x,t) & \psi 2(x,t) \\ \phi 1(x,t) & \phi 2(x,t) \end{bmatrix}$$

$$U0 = \begin{bmatrix} 0 & i\mathrm{Qs}(x,t) \\ -i\mathrm{Qs}(x,t) & 0 \end{bmatrix}$$

$$V0 = \begin{bmatrix} -\frac{1}{2}iQ(x,t)\mathrm{Qs}(x,t) & -\frac{1}{2}\mathrm{Qs}^{(1,0)}(x,t) \\ -\frac{1}{2}Q^{(1,0)}(x,t) & \frac{1}{2}iQ(x,t)\mathrm{Qs}(x,t) \end{bmatrix}$$

$$U1 = \begin{bmatrix} 1 & i0 \\ 0 & -1 \end{bmatrix}$$

$$V1 = \begin{bmatrix} 0 & -\mathrm{Qs}(x,t) \\ \mathrm{Qs}(x,t) & 0 \end{bmatrix}$$

$$V2 = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$

$$\Phi 0 = \begin{bmatrix} \psi 1(x,t) & \psi 2(x,t) \\ \phi 1(x,t) & \phi 2(x,t) \end{bmatrix}$$

$\Psi \mathrm{x} = \mathrm{U0}.\Phi + \mathrm{U1}.\Phi.\Lambda;$
$\Psi \mathrm{t} = \mathrm{V0}.\Phi + \mathrm{V1}.\Phi.\Lambda + \mathrm{V2}.\Phi.\Lambda.\Lambda;$

$\mathrm{eqx} = \mathrm{Simplify}\left[\frac{\partial \Phi}{\partial x} - \Psi \mathrm{x}\right];$

eqt = Simplify $\left[\frac{\partial \Phi}{\partial t} - \Psi t\right]$;

Simplify[eqx] // MatrixForm
Simplify[eqt] // MatrixForm

$\sigma = \Psi 0.\Lambda.\Psi 0^{-1}$;
Simplify[σ];
$\Psi N = U0 + U1.\sigma - \sigma.U1$;
Simplify[$U1.\sigma - \sigma.U1$]

$\Lambda 1 = 1$;
$\Lambda 2 = -1$;

c12 = 1;
c11 = -1;
c13 = -1;
c14 = -1;

$$V2 = \begin{bmatrix} 0 & -\frac{2c11c12(\lambda 1-\lambda 2)}{c11c14e^{\lambda 2(x-i\lambda 2t)}-c12c13e^{\lambda 1(x-i\lambda 1t)}} \\ \frac{2c13c14(\lambda 1-\lambda 2)e^{x(\lambda 1+\lambda 2)}}{c12c13e^{\lambda 1x+i\lambda 2^2t}-c11c14e^{\lambda 2x+i\lambda 1^2t}} & 0 \end{bmatrix}$$

FullSimplify[sol] // MatrixForm
$Q[x,t] := Sqrt[2]E^(It)Sech[x]$;
$Qs[x,t] := Sqrt[2]E^(-It)Sech[x]$

gpeq = $iQ^{(0,1)}(x,t) + Q^{(2,0)}(x,t) + Q(x,t)^2 Qs(x,t)$;
Simplify[gpeq]

chapterFinding peregrine solinton solution by Darboux Transformation Method

$$U0 = \begin{bmatrix} 0 & i\text{Qs}(x,t) \\ -i\text{Qs}(x,t) & 0 \end{bmatrix}$$

$$V0 = \begin{bmatrix} -\frac{1}{2}iQ(x,t)\text{Qs}(x,t) & -\frac{1}{2}\text{Qs}^{(1,0)}(x,t) \\ -\frac{1}{2}Q^{(1,0)}(x,t) & \frac{1}{2}iQ(x,t)\text{Qs}(x,t) \end{bmatrix}$$

$$U1 = \begin{bmatrix} 1 & i0 \\ 0 & -1 \end{bmatrix}$$

$$V1 = \begin{bmatrix} 0 & -\text{Qs}(x,t) \\ \text{Qs}(x,t) & 0 \end{bmatrix}$$

$$V2 = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$

Simplify[U0] // MatrixForm
Simplify[V0] // MatrixForm
Simplify[U1] // MatrixForm
Simplify[V1] // MatrixForm
Simplify[V2] // MatrixForm
Simplify[D[U0, t] - D[V0, x] + U0.V0 - V0.U0] // MatrixForm
Simplify[D[U1, t] - D[V1, x] + U0.V1 - V1.U0 + U1.V0 - V0.U1] // MatrixForm
Simplify[D[V2, x] + V2.U0 - U0.V2 + V1.U1 - U1.V1] // MatrixForm
Simplify[U1.V2 - V2.U1] // MatrixForm

$$\Lambda = \begin{bmatrix} \lambda 1 & 0 \\ 0 & \lambda 2 \end{bmatrix}$$

$$\Phi = \begin{bmatrix} \psi 1(x,t) & \psi 2(x,t) \\ \phi 1(x,t) & \phi 2(x,t) \end{bmatrix}$$

$\Psi\text{x} = \text{U0}.\Psi + \text{U1}.\Psi.\Lambda;$
$\Psi\text{t} = \text{V0}.\Psi + \text{V1}.\Psi.\Lambda + \text{V2}.\Psi.\Lambda.\Lambda;$
$\text{eqx} = \text{Simplify}\left[\frac{\partial\Psi}{\partial x} - \Psi\text{x}\right];$
$\text{eqt} = \text{Simplify}\left[\frac{\partial\Psi}{\partial t} - \Psi\text{t}\right];$
Simplify[eqx] // MatrixForm

Simplify[eqt] // MatrixForm
$Q[x,t_] := \exp(it)$
$Qs[x,t_] := \exp(-it)$
Simplify[eqx] // MatrixForm
Simplify[eqt] // MatrixForm

$\lambda 1$=1;
$\lambda 2$=-1;

$\text{eq1x} = -\lambda 1\psi 1(x,t) + \psi 1^{(1,0)}(x,t) + e^{it}\phi 1(x,t)$;

$\text{eq2x} = \lambda 1\phi 1(x,t) - e^{-it}\psi 1(x,t) + \phi 1^{(1,0)}(x,t)$;
DSolve[{eq1x = 0, eq2x = 0}, {$\phi 1(x,t), \psi 1(x,t)$}, x]

$\text{eq3x} = -\lambda 2\psi 2(x,t) + \psi 2^{(1,0)}(x,t) + e^{it}\phi 2(x,t)$;
$\text{eq4x} = \lambda 2\phi 2(x,t) - e^{-it}\psi 2(x,t) + \phi 2^{(1,0)}(x,t)$;
DSolve[{eq3x = 0, eq4x = 0}, {$\phi 2(x,t), \psi 2(x,t)$}, x]

$\phi 1(\text{x_}, \text{t_}) := (1-x)\text{c1}(t) + e^{-it}x\text{c2}(t)$;
$\phi 2(\text{x_}, \text{t_}) := (x+1)\text{c3}(t) + e^{-it}x\text{c4}(t)$;
$\psi 1(\text{x_}, \text{t_}) := (x+1)\text{c2}(t) - e^{it}x\text{c1}(t)$;
$\psi 2(\text{x_}, \text{t_}) := (1-x)\text{c4}(t) - e^{it}x\text{c3}(t)$;

Simplify[eqx] // MatrixForm
Simplify[eqt] // MatrixForm

$\text{eq1t} = -\frac{1}{2}i\left(2\lambda 1^2 + 1\right)\psi 1(x,t) + i\lambda 1 e^{it}\phi 1(x,t) + \psi 1^{(0,1)}(x,t)$;
$\text{eq2t} = \frac{1}{2}i\left(2\lambda 1^2 + 1\right)\phi 1(x,t) - i\lambda 1 e^{-it}\psi 1(x,t) + \phi 1^{(0,1)}(x,t)$;
DSolve[{eq1t = 0, eq2t = 0}, {$\phi 1(x,t), \psi 1(x,t)$}, t]
$\text{eq3t} = -\frac{1}{2}i\left(2\lambda 2^2 + 1\right)\psi 2(x,t) + i\lambda 2 e^{it}\phi 2(x,t) + \psi 2^{(0,1)}(x,t)$;
$\text{eq4t} = \frac{1}{2}i\left(2\lambda 2^2 + 1\right)\phi 2(x,t) - i\lambda 2 e^{-it}\psi 2(x,t) + \phi 2^{(0,1)}(x,t)$;
DSolve[{eq3t = 0, eq4t = 0}, {$\phi 2(x,t), \psi 2(x,t)$}, t]
Simplify[eqx] // MatrixForm
Simplify[eqt] // MatrixForm
$\text{eq1t} = \psi 2^{(0,1)}(x,t) - \frac{1}{2}i\lambda 2\psi 2(x,t)\left(\lambda 2\text{a1}(t)(\text{f11} - \text{f110})^2 - 2i\text{g11}\right)$;
$\text{eq2t} = \frac{1}{2}\left(2\phi 2^{(0,1)}(x,t) + \lambda 2\phi 2(x,t)(-2\text{g11} + \text{a1}(t)(\text{f11} - \text{f110})(4\text{c4}(t) + i\lambda 2(\text{f11} - \text{f110})))\right)$;
Simplify[eq1x]

Simplify[eq2x]
Simplify[eq1t]
Simplify[eq2t]

DSolve[eq1x=0,ψ2(x,t),x]
DSolve[eq2x=0,ϕ2(x,t),x]

$\left(\psi 2(\mathrm{x_},\mathrm{t_}):=\mathrm{c111}(t)e^{\mathrm{f11}\lambda 2x}\right)$
$\left(\phi 2(\mathrm{x_},\mathrm{t_}):=\mathrm{c112}(t)e^{\mathrm{f110}\lambda 2x}\right)$
Simplify[eq1x]
Simplify[eq2x]
Simplify[eq1t]
Simplify[eq2t]

DSolve[eq1t == 0, c111[t], t]
DSolve[eq2t == 0, c112[t], t]

$\mathrm{c111}(\mathrm{t_}):=\mathrm{c1110}\exp\left(\int \frac{1}{2}\left(i\mathrm{f11}^2\lambda 2^2\mathrm{a1}(t)-2i\mathrm{f11f110}\lambda 2^2\mathrm{a1}(t)+i\mathrm{f110}^2\lambda 2^2\mathrm{a1}(t)+2\mathrm{g11}\lambda 2\right)\,dt\right)$
$\mathrm{c112}(\mathrm{t_}):=\mathrm{c1120}\exp\left(\int \frac{1}{2}\left(-4\mathrm{f11}\lambda 2\mathrm{a1}(t)\mathrm{c4}(t)+4\mathrm{f110}\lambda 2\mathrm{a1}(t)\mathrm{c4}(t)-i\mathrm{f11}^2\lambda 2^2\mathrm{a1}(t)+2i\mathrm{f11f110}\lambda 2^$

$Simplify[\Psi 2[x,t]]$
$Simplify[\Phi 2[x,t]]$
Simplify[eq1x]
Simplify[eq2x]
Simplify[eq1t]
Simplify[eq2t]

$\phi 1\mathrm{s}(x,t)=\mathrm{Simplify}[\psi 2(x,t)]$
$\psi 1\mathrm{s}(x,t)=\mathrm{Simplify}[-\phi 2(x,t)]$

r=1;
s=1;
$\left(\psi 1(\mathrm{x_},\mathrm{t_}):=-\mathrm{c1120}\exp\left(\mathrm{f110}\lambda 2x+\frac{1}{2}\lambda 2(\int(2\mathrm{g11}+i\mathrm{a1}(t)(\mathrm{f11}-\mathrm{f110})(\lambda 2(\mathrm{f11}-\mathrm{f110})+4i\mathrm{c4}(t))\right.\right.$
$\psi 2(\mathrm{x_},\mathrm{t_}):=\mathrm{c1110}\exp\left(\mathrm{f11}\lambda 2x+\frac{1}{2}i\lambda 2\left(\int(\lambda 2\mathrm{a1}(t)(\mathrm{f11}-\mathrm{f110})^2-2i\mathrm{g11})\,dt\right)\right)$
$\phi 1(\mathrm{x_},\mathrm{t_}):=\mathrm{c1110}\exp\left(\mathrm{f11}\lambda 2x-\frac{1}{2}i\lambda 2\int(\lambda 2\mathrm{a1}(t)(\mathrm{f11}-\mathrm{f110})^2+2i\mathrm{g11})\,dt\right)$
$\phi 2(\mathrm{x_},\mathrm{t_}):=\mathrm{c1120}\exp\left(\mathrm{f110}\lambda 2x+\frac{1}{2}\lambda 2(\int(2\mathrm{g11}-i\mathrm{a1}(t)(\mathrm{f11}-\mathrm{f110})(\lambda 2(\mathrm{f11}-\mathrm{f110})-4i\mathrm{c4}(t)))\,dt\right)$
r = 1;

s = 1;

c2(t_):=$i\sqrt{\frac{\text{a2}(t)}{2\text{a1}(t)}}$

$$U0 = \begin{bmatrix} 0 & -\frac{\sqrt{\frac{\text{a2}(t)}{\text{a1}(t)}}e^{ix\text{c4}(t)}Q(x,t)}{\sqrt{2}} \\ \frac{\sqrt{\frac{\text{a2}(t)}{\text{a1}(t)}}e^{-ix\text{c4}(t)}\text{Qs}(x,t)}{\sqrt{2}} & 0 \end{bmatrix}$$

$$U0 = \begin{bmatrix} f11 & 0 \\ 0 & f110 \end{bmatrix}$$

$$\Lambda = \begin{bmatrix} \lambda 1 & 0 \\ 0 & \lambda 2 \end{bmatrix}$$

$$\Phi 0 = \begin{bmatrix} \psi 1(x,t) & \psi 2(x,t) \\ \phi 1(x,t) & \phi 2(x,t) \end{bmatrix}$$

Simplify[σ];
ΨN = U0 + U1.σ − σ.U1;
ΨN = U0 + U1.σ − σ.U1;
Simplify[ΨN]
Simplify[U1.σ − σ.U1]

f11=1;
f110=2;
g11=0;
a2(t_):=1
a1(t_):=2;
c4(t_):=-t
Simplify[Q(x,t)] Simplify[Q(x,t) Qs(x,t)]

ψsq(x_, t_):=$\frac{8\text{a1}(t)(\text{f11}-\text{f110})^2\exp\left(\frac{1}{2}\left(\int(2\text{g11}+i\text{a1}(t)(\text{f11}-\text{f110})(4i\text{c4}(t)+\text{f11}-\text{f110}))\,dt+\int(2\text{g11}+i\text{a1}(t)(\text{f11}-\text{f110})(4i\text{c4}(t)+\right.\right.}{\text{a2}(t)\left(\exp\left(\frac{1}{2}\left(\int(2\text{g11}+i\text{a1}(t)(\text{f11}-\text{f110})(4i\text{c4}(t)+\text{f11}-\text{f110}))\,dt+\int(2\text{g11}+i\text{a1}(t)(\text{f11}-\text{f110})(4i\text{c4}(t)-\text{f11}+\text{f110}\right.\right.\right.}$

$Simplify[\Psi sq[x,t]]$ $Q[x,t_] = -\frac{4e^{2t^2+it(x+2)+x}}{e^{4t^2}+e^{2x}}$